

Affordability, Financial Innovation, and the Start of the Housing Boom *

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Abstract

Non-traditional mortgage features are the subject of extensive regulatory and popular criticism for their role in the housing boom and bust. In this paper, we construct a novel county-level dataset to analyze how mortgage products transmitted the mid-2000s credit supply shock to local markets. Using quasi-experimental spatial variation in exposure to the private label securities (PLS) market and the heterogeneous timing of housing booms, we provide new evidence that non-traditional mortgage products allowed lenders to expand credit in an environment with rising interest rates and house prices by maintaining payment affordability, thereby fueling house price growth.

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1 Introduction

The 30-year fixed-rate mortgage (FRM) is the dominant product in the United States housing market, comprising over three-quarters of mortgage originations.¹ However, this standard mortgage became significantly more expensive with the Federal Reserve’s interest rate hikes amid inflationary pressures. When mortgage rates doubled between December 2021 and a year later, the monthly payment on a median-priced home jumped from \$1,449 to \$2,140, a nearly 50 percent increase.² Such a rise in costs makes alternatives to the standard mortgage more attractive to borrowers, especially without corresponding declines in house prices.

To relax a binding payment constraint, borrowers have historically turned to mortgages with lower initial rates, such as traditional adjustable-rate mortgages (ARMs). However, “non-traditional” or “alternative” mortgage products were the preferred alternative in the mid-2000s. These products included mortgages with introductory teaser rates, interest-only or negative amortization periods, extended payment terms, and deferred balloon payments. Illustrated in Figure 1, the popularity of these products surged during the housing boom when both interest rates and house price growth were on the rise. This coincidence led the Financial Crises Inquiry Commission to conclude that what they called “toxic” mortgage products were a cause of the housing boom.³ As a result, the Qualifying Mortgage (QM) rule issued by the Consumer Financial Protection Bureau now excludes these products, making them ineligible for sale to the Government Sponsored Enterprises (GSEs).

Today’s housing market mirrors the housing boom of the 2000s in combining rising house prices with high interest rates. Despite a noticeable decline in housing affordability since 2021, the QM rule effectively restricts non-traditional mortgages to a minor role in the non-agency market. In 2023, only 2.7% of first-lien purchase mortgages had at least one alternative financing feature.⁴ Thus, the role of non-traditional mortgage products during the mid-2000s remains a pivotal question. If these products were more symptom than cause, then the stringent regulations that now govern mortgage contracts might unnecessarily restrict access to affordable financing options in today’s high-interest-rate environment (Cocco 2013). Answering this question requires understanding precisely how non-traditional products fit into the broader credit expansion that preceded the crisis.

A large body of research has established that an expansion in mortgage credit supply was a primary driver of the mid-2000s housing boom (Mian and Sufi 2009, 2021; Favilukis et al. 2013;

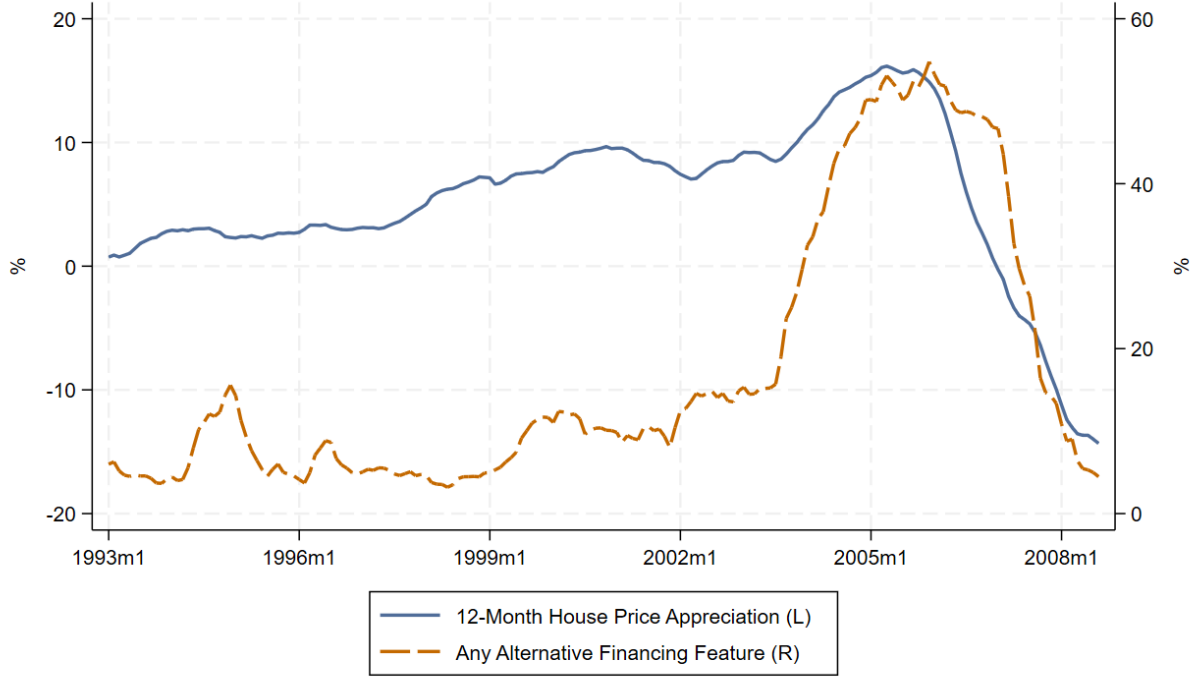
¹Urban Institute Housing Finance Chartbook (2024).

²Based on the 2021q4 median home price of \$423,600 reported by the Census, loan-to-value of 80 percent, and average mortgage rates of 3.11 and 6.49 for December 2021 and 2022, respectively, from Freddie Mac’s Primary Mortgage Market Survey.

³The Financial Crisis Inquiry Report (2011).

⁴Home Mortgage Disclosure Act (2024).

Figure 1: National Trends in Alternative Financing, Payments, and Prices



Note: This figure shows the share of purchase loans originated in a given month with any alternative financing feature and 12-month house price appreciation.

Favilukis, Ludvigson and Nieuwerburgh 2016; Nadauld and Sherlund 2013; Griffin, Kruger and Maturana 2021; Justiniano, Primiceri and Tambalotti 2022). Yet, this literature is largely agnostic about the specific channels through which credit expansion translated into higher house prices. In practice, financial liberalization is commonly modeled as an opaque bundle of regulatory and market changes, leaving open whether the operative margin was the relaxation of documentation standards, the expansion of subprime lending, or the proliferation of new mortgage contract structures.

In this paper, we investigate the role of non-traditional mortgages in facilitating credit expansion through reduced mortgage payments and their impact on house price growth. We construct a novel and comprehensive county-by-month panel dataset comprising over 60 million mortgage loans. To measure changes in payment affordability through contract features, we introduce an innovative analytical tool we term the “payment gap” that quantifies the disparity between actual monthly mortgage payments and counterfactual payments under a standard 30-year FRM. Our analysis of the impact of non-traditional mortgages on affordability and house prices then employs two complementary empirical approaches: first, leveraging an exogenous shock to credit supply in

an instrumental variables specification and, second, using an event-study design around the start of local housing booms.

Our first approach leverages the rapid expansion of the Private-Label Securitization (PLS) market in response to Treasury rate increases in mid-2003 (Justiniano, Primiceri and Tambalotti 2022). As Mian and Sufi (2021) show, these rate increases created plausibly exogenous variation in credit supply relative to local mortgage demand, driven by differential exposure to lenders financing mortgages through the PLS market. They find that after the rate increases, house prices rose substantially more in markets with high ex ante concentrations of such lenders. Changes in product choice are a plausible mechanism, as non-traditional mortgages were almost exclusively offered by lenders securitizing loans through the PLS market. We use the PLS expansion as an instrument for each margin of credit supply to assess for which dimensions of credit supply it serves as a relevant instrument. To complement these IV estimates, we provide difference-in-differences estimates comparing counties across quartiles of exposure to lenders in the PLS market to examine the relative timing of changes across credit supply dimensions.

This approach yields new evidence that non-traditional mortgage products are a primary mechanism that transmitted the 2003 credit shock to housing markets. The instrumental variables specification suggests that the expansion of the PLS market had a strong impact on product composition, especially interest-only and negatively amortizing mortgages.⁵ Immediately following the rate increase, use of non-traditional mortgage products differentially increased sharply and immediately in counties with more exposure to securitization. Within six months, counties with the most exposure differentially increased use of non-traditional mortgages by 5.0 percentage points compared to the least exposed. By twelve months, this difference expanded to 11.5 percentage points. As a result, we estimate that the median monthly mortgage payment gap across all loans declined by an average of \$72 more in counties with the most exposure to securitization relative to counties with the least exposure until the end of the boom.

Importantly, these product and payment differences appear *before* differences in house price appreciation. The payment gap subsequently kept growth in median realized mortgage payments in these counties on pace with other counties until late in the boom period, despite much more rapid house price growth. Other mortgage features, such as occupancy status and subprime borrowers, widely associated with the expansion of the PLS market and deteriorating loan quality also concurrently change. However, these features have a weaker association with the expansion and respond more slowly than product composition.

⁵This aggregate pattern is consistent with the shift by one large subprime lender (New Century) toward interest-only loans in 2004 in response to the change in monetary policy (Landier, Sraer and Thesmar 2015).

In our second empirical approach, we explore the relationship between non-traditional mortgage products and rising house prices outside the specific mid-2003 setting to paint a fuller picture of the impact of their use. To do so, we first identify the onset of local house price booms dating back to the early 1990s and then conduct an event study that examines the time-series relationship between the start of these booms and the use of non-traditional products and other measures of credit supply. We define the start of a housing boom as the month in which 12-month house price appreciation deviates most sharply from prior trends. This month is identified using a county-specific breakpoint randomization methodology, based on the theory in [Hansen \(2000\)](#) and similar to that used by [Charles, Hurst and Notowidigdo \(2018\)](#) and [Ferreira and Gyourko \(2023\)](#). The analysis is conducted separately for booms starting before and after the expansion of the PLS market starting in 2002.

This approach produces findings for late booming markets that are strikingly consistent with the patterns observed after the 2003 Treasury rate shock. However, our findings indicate that non-traditional products were an unlikely contributor to earlier housing booms. In markets with early housing booms before February 2002, the preference for traditional ARMs to maintain affordability as house prices rose is the most notable feature. During this period, non-traditional mortgage use and the payment savings they offered declined modestly and steadily around the start of housing booms. At the same time, other indicators of credit supply signaled a tightening of credit conditions through the start of local booms. These findings for early booms challenge the notion that products offering greater affordability than traditional ARM products are intrinsically detrimental to housing markets. The lesson instead may be that these products become problematic when their complexity helps obscure risks to investors when credit is extended to increasingly speculative and subprime borrowers.

In the future, non-traditional mortgage restrictions may ease or emerging financial structures may similarly relax monthly mortgage payment constraints. For instance, an important recent development is homebuilders offering temporary mortgage rate “buydowns” through lender subsidiaries, effectively recreating teaser mortgages.⁶ The payment gap measure we develop is a valuable tool for broad monitoring of mortgage credit conditions amid such complexity. Unlike standard measures of affordability, such as loan-to-value, debt-to-income, and payment-to-income, the payment gap allows for a comparison of affordability by holding income, loan amount, and prices constant while varying contract type. This distinction is crucial when borrowers have the option to choose among different mortgage products. By analyzing variation in the payment gap across different borrower groups, current and future policymakers can gain deeper insight into credit conditions along a critical margin influencing mortgage decisions.

⁶A recent industry survey showed that 75% of nationally surveyed homebuilders are offering mortgage rate buydowns on new homes ([John Burns Research & Consulting \(2024\)](#)).

Our work aligns with the broader literature investigating the role of credit expansion in house price dynamics during the boom ([Glaeser, Gottlieb and Gyourko 2013](#); [Landvoigt, Piazzesi and Schneider 2015](#)). Previous studies have utilized exogenous variation in credit supply through conforming loan limits ([Adelino, Schoar and Severino 2025](#)), state branching regulations ([Favara and Imbs 2015](#)), and predatory lending laws ([Di Maggio and Kermani 2017](#)). We follow [Mian and Sufi \(2021\)](#) and [Justiniano, Primiceri and Tambalotti \(2022\)](#) in using the 2003 Treasury rate increase and concurrent expansion of the PLS market to isolate variation in credit supply. By employing credible identification, we quantify how non-traditional mortgage use expanded credit availability and anticipated house price accelerations during the early 2000s. Moreover, our event study across housing cycles suggests the pivotal issue in the 2000s was the deployment of non-traditional mortgage products by the PLS market, rather than an inherent tendency of non-traditional products to generate housing booms.

We analyze a broad range of non-traditional mortgage features appealing to both prime and subprime borrowers, emphasizing their distinct role in driving house price increases. Previous research has focused on the theoretical and empirical characteristics of these contracts and the borrowers who use them. For example, [Amromin et al. \(2018\)](#) and [Keys et al. \(2013\)](#) highlight how high-income borrowers with strong credit profiles preferred interest-only and negative-amortization mortgages during the boom, whereas teaser-rate loans were more common in the subprime market. [Piskorski and Tchisty \(2010\)](#) and [Piskorski and Tchisty \(2011\)](#) find that optimal mortgage contracts in rising markets mirror those popular in the boom. [Barlevy and Fisher \(2011\)](#) demonstrate that interest-only mortgages can spur speculative bubbles and show that their increased use anticipated house price rises. [Brueckner, Calem and Nakamura \(2016\)](#) argue high house price expectations increased the use of interest-only and negatively amortizing mortgages to lower default risk. Our analysis extends these findings by studying how non-traditional mortgages were deployed to increase payment affordability and impact house prices during the boom. A related body of research studies the impact of alternative mortgage contracts on the severity of the subprime crises once prices began to fall ([Gerardi et al. 2008](#); [Demyanyk and Van Hemert 2011](#); [Palmer forthcoming](#)).

These results integrate a number of strands of literature focused on the degree to which the housing boom reflected price increases that outpaced underlying fundamentals ([Himmelberg, Mayer and Sinai 2005](#); [Coleman, LaCour-Little and Vandell 2008](#); [Mian and Sufi 2009](#); [Davidoff 2013](#); [Adelino, Schoar and Severino 2016](#); [Ferreira and Gyourko 2023](#); [Chodorow-Reich, Guren and McQuade 2024](#)). Indeed, a number of papers point to the role of investors in driving speculation ([Haughwout et al. 2011](#); [Chinco and Mayer 2016](#); [DeFusco, Nathanson and Zwick 2022](#); [Mian and Sufi 2021](#)). This debate remains unsettled, in part because of the challenge of capturing how

expectations of future house price increases affect borrowers' and lenders' mortgage origination choices. Our evidence supports the view that the increased use of non-traditional mortgage products was untethered to meaningful improvements in local fundamentals in later and more severely booming markets and aided speculative purchases.

2 Data and Measurement

The dataset we construct has three key features that make it uniquely well-suited to studying the relationship between non-traditional mortgage products and rising house prices. First, we are able to observe the house price dynamics of a large number of local housing markets. Importantly, our repeat-sales house price indices are based on sales backed by a large range of loan types rather than just conforming loans. These indices should capture the house price movements of properties financed with non-traditional mortgage products, which is likely to reduce measurement error. Second, we take advantage of administrative loan-level data that is comprehensive in its geographic and loan characteristic coverage. This data allows us to measure with precision the prevalence of non-traditional mortgage products and payments for a large number of local markets back to the early 1990s.⁷ To our knowledge we are the first to construct these measures. Third, using multiple years of administrative loan-level data allows us to take measurements at a monthly frequency, which makes it possible to pinpoint sharp changes in our variables and to follow the order in which they occur. In summary, our dataset provides more granularity across three dimensions – the geographic level (county), time period (month), and mortgage-level detail – than any previous study along these lines. We now discuss each data source in more detail.

The county-level house price indices are from CoreLogic (CL). These indices are, for our purposes, preferable to other indices, such as those produced by the Federal Housing Finance Agency (which only include properties backed by conforming loans), or those produced by Case-Shiller (which are limited to a smaller set of markets). At the time of our analysis, CL produced 1,163 county-level house price indices for single-family detached properties using their repeat sales methodology. CL indices have the advantage of going back for many more years than other indices, such as Zillow (though there is considerable overlap between the two sources in recent years). The CL indices start in 1973 for some counties, but we begin our analysis in 1993, when many indices begin. No indices are produced for counties that experience fewer than five sales in

⁷Non-traditional mortgage products came into use with the emergence of ARMs in the 1980s (Peek 1990). In the late 1990s, their use broadened, as they became more attractive to lenders for several reasons. For one, advances in technology, including automated underwriting and better credit monitoring, made lenders more confident in their ability to assess the risk of such products (Gates, Perry and Zorn 2002). In addition, in a healthy economy with rising house prices, lenders may have been more willing to make contracts with incomplete information (Bolton and Faure-Grimaud 2010; Tirole 2009; Piskorski and Tchistyi 2011).

a least one month over their collection period. The sample counties cover over 80 percent of the US population. The remaining 20 percent largely consists of rural counties that experienced very little of the housing boom.

Our mortgage characteristics are from the Home Mortgage Disclosure Act (HMDA), CoreLogic, and Black Knight McDash. Passed by Congress in 1975, HMDA requires every lender satisfying any of a broad list of criteria to report every loan and a set of its characteristics to a central repository.⁸ Because of this requirement, coverage for this dataset is broad. CL and Black Knight McDash provide loan-level records for non-agency (mostly subprime and alt-A) mortgages and agency (mostly prime) mortgages, respectively. Both CL and Black Knight McDash collect these records from a group of mortgage servicers. We limit the data from each source to purchase loans since we are interested in the actions of borrowers entering the market, rather than current owners who refinance or extract equity from their homes.

The combined coverage by CL and Black Knight McDash of the first-lien purchase market is less complete than HMDA. The servicer-based data coverage in the 1990s is roughly 30 percent of HMDA and grows with the addition of more servicers over time. At the height of the housing boom, these two sources cover loans comparable to three-quarters of the first-lien market captured by HMDA.⁹ While the representativeness of our data in earlier years and compositional shifts over time are a concern, this data remains the best, if not only, source for detailed information on loan characteristics. Additionally, for the purposes of our analysis, it is unlikely that changes in coverage are correlated with the timing of changes in monetary policy, county-level structural breaks in house prices, or our other market variables.

Figure 2 shows the prevalence over time of each alternative financing feature we measure using the combined CL and Black Knight McDash servicer data. These features include the percent of loans with an extended term longer than 30 years, a balloon payment, an allowed negative amortization period, an interest only period, and a teaser rate, which we define as an initial fixed-rate term lasting more than one and up to three years.^{10,11} These categories are not exclusive, as loans can have multiple alternative financing features. For example, an Option ARM can both allow for

⁸The most important criteria include having assets above a low threshold and operating a branch within an MSA. For a full discussion of reporting requirements see [Federal Financial Institutions Examination Council \(2013\)](#).

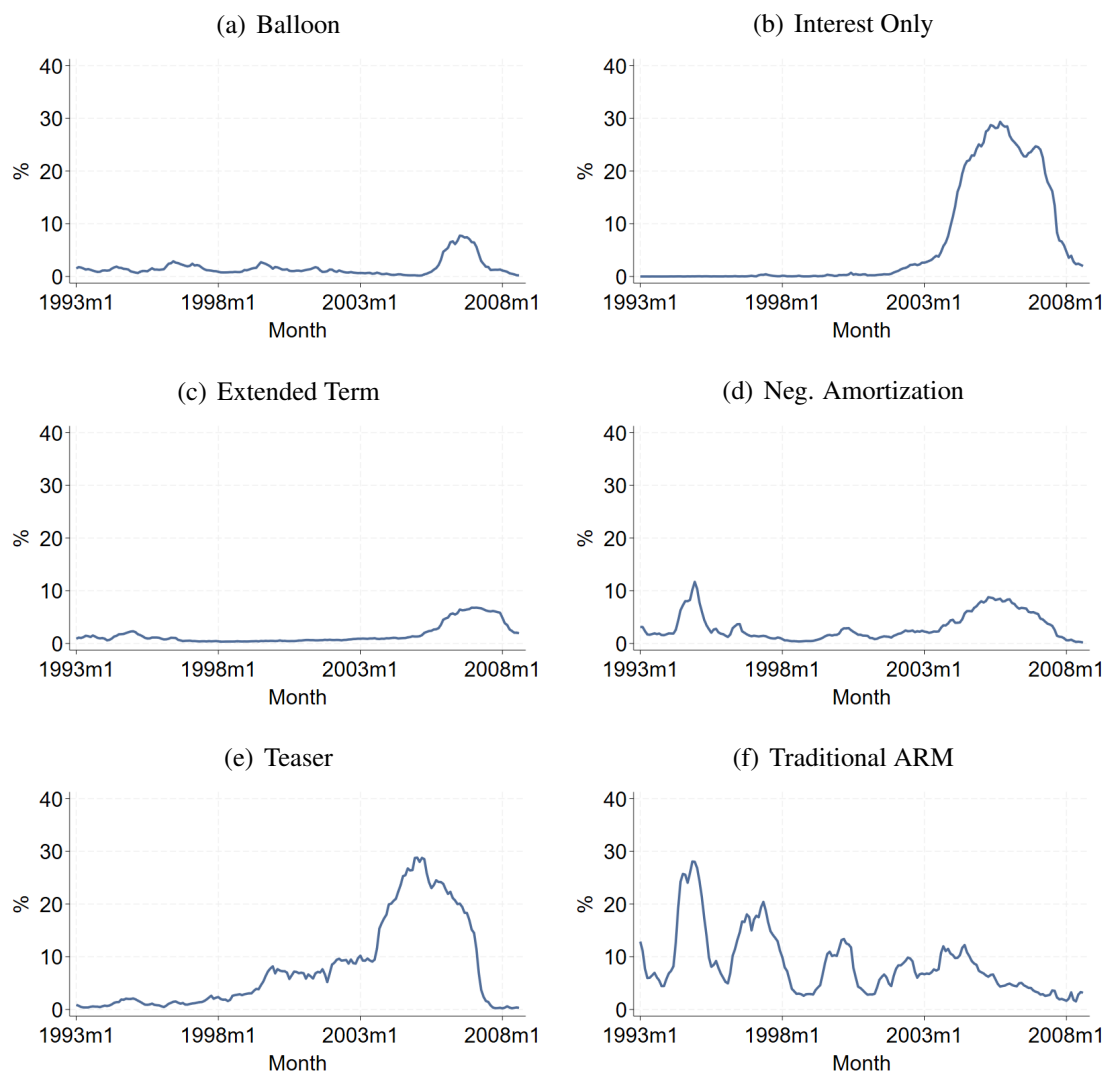
⁹Figure A1 shows the coverage of CL and Black Knight McDash over time in loan counts and volumes as compared to HMDA.

¹⁰As discussed in [Man \(2004\)](#), other types of non-traditional mortgage products were available before 1990, including the price level adjusted mortgage, shared appreciation mortgage and graduated payment mortgage. However, these were never widely used.

¹¹Note that not all loans with allowed negative amortization in CL are ARMs. For example, we observe a small minority of standard 30-year FRMs from this source with a negative amortization flag because payments that are insufficient to cover the interest owed can result in uncovered interest being added to the principal. All loans in Black Knight McDash categorized with allowed negative amortization are Option ARMs.

negative amortization and carry an introductory teaser rate. For comparison, we also calculate the percent of loans with more traditional ARM structures, an ARM without any alternative financing feature.

Figure 2: Alternative Financing Features



Note: This figure shows the percent of purchase loans originated with specific alternative financing features. Loans can have more than one alternative financing feature. Traditional ARMs are shown for comparison.

There are several notable features in the aggregate time series. First, non-traditional mortgage products were in general use since the early 1990s, though their use was limited. Second, in the late 1990s, loans with initial teaser rates jumped to 7 percent of all loans. This pattern is consistent with the emergence of a nascent subprime market that predated the more notorious version of the mid-2000s. Third, there is a strong correlation in the market share of traditional ARM products and periods of rising interest rates prior to the 2000s. However, during the early years of the

housing boom of the 2000s, traditional ARMs were largely displaced by non-traditional mortgages, particularly interest-only loans and loans with teaser rates. Finally, extended term and balloon payment features rise later in the boom. These features were often combined with interest rate-modifying features.¹²

In addition to the prevalence of non-traditional products and mortgage payments, we examine other borrower and mortgage attributes that capture additional margins along which lenders could expand credit supply. From the CL and Black Knight McDash data, these include the average first-lien loan-to-value (LTV) ratio, average credit (FICO) score at origination, and percent of first-liens with full documentation. The measures we calculate from HMDA include the percent of loan applications denied, percent owner-occupied, second lien percent of originations, percent FHA, and percent subprime. Apart from a higher prevalence of low-LTV and high FICO loans at the start of the sample, likely reflecting a compositional shift in covered servicers at the beginning of our sample, the series match the known features of mortgage originations over time. Additional discussion of these features can be found in Appendix Section 6.A.

To build our panel, we calculate the percent of loans with each individual alternative financing feature and traditional ARMs and summary measures of other borrower and mortgage attributes for each county and month from 1993-2010. To this we add summary measures of payments and payment affordability. County household income and unemployment rates are obtained from the Census and the Bureau of Labor Statistics (BLS) to track changing local economic fundamentals.

2.A Payment Gap

Throughout our analysis, we investigate the influence of changing mortgage product composition on monthly mortgage payments at origination by analyzing the disparity between actual contracted payments and counterfactual payments under a standard 30-year FRM. This counterfactual is constructed by first estimating the relationship between the spread on mortgage contract rates over the 10-year Treasury rate and key loan characteristics for all 30-year FRMs originated in 1995. Our model follows:

$$Rate_{icm} - TRate_m = \sum_{f=1}^5 \sum_{l=1}^5 \beta^{fl} FICO_i^f LTV_i^l + \phi_c + \mu_{icm}, \quad (1)$$

where $Rate_{icm}$ is the mortgage rate of borrower i in county c during month m , $TRate_m$ is the concurrent 10-year Treasury rate, $FICO_i^f$ and LTV_i^l denote dummies for the borrower's FICO score and LTV ratio at origination falling into one of five buckets, respectively, in line with the GSE's

¹²Appendix Figure A2 shows that half of extended term and balloon payment mortgages were combined with interest-modifying features from 2004–2007.

loan-level price adjustment matrix, and ϕ_c are county-specific fixed effects.¹³ Then, utilizing this model, we project the counterfactual interest rate for 30-year FRMs for all loans originated during our sample period. This projection incorporates the progression of the 10-year Treasury rate and coefficients from equation 1:

$$\widehat{Rate}_{icm} = TRate_m + \sum_{f=1}^5 \sum_{l=1}^5 \widehat{\beta}^{fl} FICO_i^f LTV_i^l + \widehat{\phi}_c. \quad (2)$$

The estimated payment for a hypothetical 30-year FRM, calculated using $\widehat{Rate}_{icm}$, provides an estimate of the payment a borrower could have expected on a standard mortgage if pre-boom (1995) lending conditions had persisted. Constructing this counterfactual requires several simplifying assumptions. First, we assume that FICO scores and LTV ratios would continue to impact interest rates similarly over time. Second, we assume stability in the spatial distribution of credit liquidity across counties over time. Third, we assume that lenders apply this same pricing model across mortgage products.

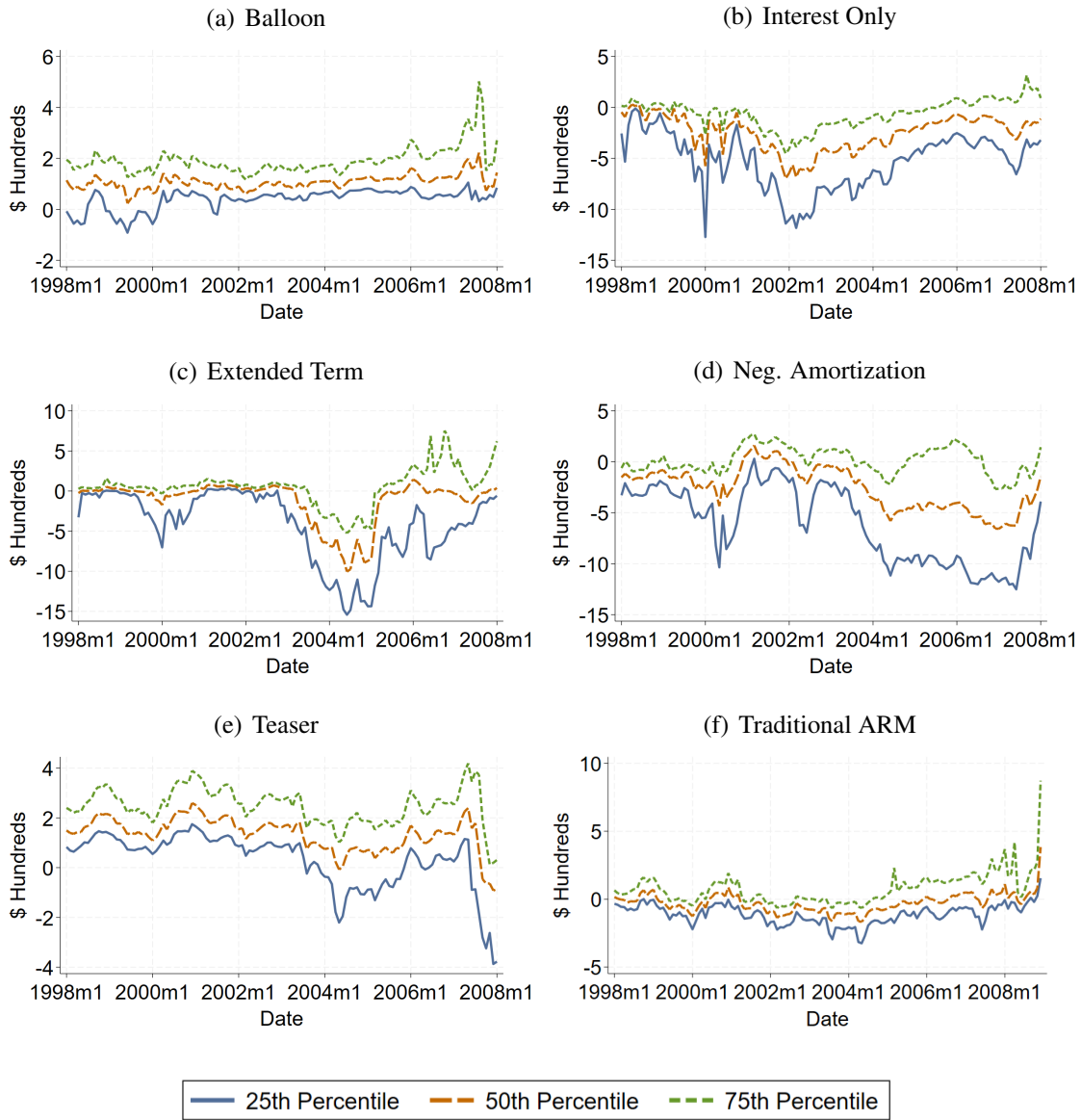
Our approach is conservative in estimating counterfactual payment savings. We hold the spread between 10-year Treasury rates and 30-year FRM mortgage rates constant at its 1995 level. In practice, the Freddie Mac Primary Mortgage Market Survey shows that mortgage rates rose relative to Treasuries after 1995, so our counterfactual estimate is downward biased. In addition, for many subprime borrowers a more realistic counterfactual would be exclusion from the mortgage market altogether, reinforcing this downward bias. Finally, mortgages with terms shorter than 30 years produce counterfactual payments that are typically higher than actual payments. This is the case for many balloon mortgages.

We define the “payment gap” as the discrepancy between the contracted monthly mortgage payments and our counterfactual estimates. A negative payment gap indicates that the actual mortgage payment provides savings relative to its counterfactual. While the level of the aggregate payment gap at one point in time is informative, its underlying assumptions and conservative estimation imply the primary value of this measure is in its changes over time. We explore this time series variation in our empirical approaches.

Figure 3 shows the interquartile range of the payment gap for loans with different alternative financing features. Negative payment gaps are measured for more than three-quarters of loans with interest-only, negative-amortization, and extended terms. At the height of the housing boom in April 2005, the most popular interest-only loans offered monthly savings of more than \$169 to half of such mortgages and more than \$387 to a quarter. Negatively amortizing and extended-term

¹³Coefficients, shown in Appendix Table A1, show the expected relationship that borrowers with lower FICO scores and higher LTVs generally receive higher mortgage rates.

Figure 3: National Trends in the Payment Gap for Loans with Alternative Financing Features



Note: The figure shows the 25th, 50th, and 75th percentiles of the payment gap between the actual mortgage payment and counterfactual 30-year FRM payment for loans with each of five alternative financing features. Traditional ARMs are shown for comparison. A negative payment gap indicates that a loan has a smaller mortgage payment than the counterfactual payment. For the individual features, we only show payment gaps starting in 1998 because the rarity of specific loan types creates uninformative extreme values in some earlier periods.

mortgages offered even larger savings to those that received them, in excess of \$873 and \$571, respectively for a quarter of such borrowers. For both balloon and subprime mortgages there are declines in the size of the payment gap across the distribution during the mid-2000s, with the more substantial affordability gains at the bottom of the distribution. For example, the 25th percentile

of the payment gap for teaser loans declined by \$306 from January 2002 to May 2004.¹⁴ In sum, alternatives to the 30-year FRM allowed borrowers to avoid paying thousands of dollars in mortgage payments during the housing boom, with substantial heterogeneity over time and across financing features.

3 Evidence from the 2003 Treasury Rate Increase

In our first approach to understanding the role of non-traditional mortgages in the interaction between credit supply and house prices, we focus on a discrete, plausibly exogenous supply-side change during an era of broad financial liberalization: the sharp Treasury rate increases in mid-2003 that fueled a rapid expansion in the PLS market (Justiniano, Primiceri and Tambalotti 2022). We use the fact that non-core deposits reliably characterize ties to the PLS market and that these funding structures were highly persistent over time, allowing pre-shock exposure to capture the impact of the PLS market expansion across geography (Mian and Sufi 2021). This expansion led to aggregate credit supply growth through changes in how lenders financed their mortgage business. Lenders traditionally relied on deposits and other core liabilities to fund mortgages, holding the mortgages on balance sheet. Alternatively, lenders tied to the PLS market tended to follow an originate-to-distribute model in which selling loans to the secondary market obviated the need for substantial core liabilities. Indeed, many non-bank lenders exclusively securitize all mortgages originated.

This sharp expansion of the PLS market in 2003 also acted as an exogenous shock to the menu of mortgage contracts offered to borrowers. Non-traditional mortgage products were almost exclusively securitized through the PLS market in the early 2000s due to the conforming loan standards for securitization set by the GSEs. For example, in 2004, among the portfolio of loans held by Freddie Mac and Fannie Mae, 82 and 88 percent, respectively, were intermediate- to long-term FRMs. At the same time, interest-only and negatively amortizing adjustable-rate loans accounted for only 2 and 3 percent of total loans, respectively. Judging these loans to be of higher risk, the GSEs never made direct purchases of such loans a core part of their business (though the GSEs were more exposed to this risk by the securities they purchased from the PLS market).¹⁵ By re-

¹⁴Additional summary statistics for the payment gap are available in Appendix Section 6.B.

¹⁵Freddie Mac intermediate-term loans start at terms of 15 years, while those for Fannie Mae start at 20 years. Both GSEs also purchased a small amount of loans with balloon payments in 2004. After 2004, the GSEs modestly increased their purchases of non-traditional mortgages in response to the higher prevalence of such mortgages in the market and competition from the PLS market. As a result, for example, about 6 percent of total loans on book for Fannie Mae in 2005 and 2006 were interest-only or negative amortizing ARMs. For more information, see the annual reports for Freddie Mac (2006) and Fannie Mae (2005 and 2006).

ducing monthly payment constraints, these non-traditional mortgages could facilitate the aggregate credit expansion.

3.A Exposure to PLS Market Event Study Design

We extend the methodology from [Mian and Sufi \(2021\)](#) to study the impact of the expansion of the PLS market on alternative financing features against changes in other mortgage attributes. To do so, we first measure the share of each lender’s liabilities that are non-core deposits in 2002, $NCL_{j,2002}$, where NCL denotes non-core liabilities and j indexes lenders.¹⁶ County-level exposure to the PLS market, $NCL_{c,2002}$ is then defined as the sum of lender non-core deposit shares weighted by the lender’s dollar value share of mortgage originations in the county in 2002, $w_{cj,2002}$:

$$NCL_{c,2002} = \sum_{j \in c, 2002} w_{cj,2002} NCL_{j,2002}. \quad (3)$$

Our empirical approach is to measure the relationship between different credit supply indicators and cumulative house price growth during the boom across counties, employing exposure to non-core lenders as an instrumental variable. We use a bivariate instrumental variable model,

$$CHPA_c = \eta_0 + \eta_1 X_c + \mu_c, \quad (4)$$

where $CHPA_c$ represents cumulative house price appreciation from January 2002 to April 2005, and X_c denotes a credit supply measure, instrumented by $NCL_{c,2002}$. We utilize standardized values of X_c .

A key distinction from [Mian and Sufi \(2021\)](#) is that their approach regresses outcomes directly on NCL exposure in a reduced form specification. Our approach instead uses NCL exposure as an instrumental variable for specific measures of credit supply. This allows us to study the first-stage relationships between NCL exposure and these credit supply measures. We can therefore ask not just whether NCL exposure causally affected house prices, which their work establishes, but through which dimensions of credit supply that effect operated. An important caveat to this exercise is that it does not delineate the independent causal impact of each credit supply dimension on house prices. Such an exercise would require multiple instruments.

We also compare changes in outcomes across counties in each of the top three quartiles of $NCL_{c,2002}$ against the bottom quartile over time in a difference-in-differences specification. We study differences at a quarterly frequency. Denote quarters as relative to $q_0 = 2003q2$ and $D_q^l =$

¹⁶To construct this measure of exposure, we merge HMDA data with banks’ Call Report data from 2002. We thank Neil Bhutta for providing his lender-ID crosswalk across these datasets.

$\mathbb{1}\{q - q_0 = l\}$ as the relative quarter indicator. Our specification is

$$Y_{cm} = \sum_{Q=2}^4 \sum_{q=-12, q \neq 0}^{16} \alpha^{Ql} D_q^l NCL_{c,2002}^Q + \mu_{cm} \quad (5)$$

where Y_{cm} is an outcome for county c and time m and the $NCL_{c,2002}^Q$ are dummies for each county belonging to one of the top three quartiles, Q , of NCL exposure. The α^{Ql} measure the differential change in each quartile relative to the first.

This specification complements the IV analysis. The timing of changes in housing market features and house prices relative to the 2003 rate shock provides additional evidence on the credit supply channel through which the PLS market operated. The specification also provides supporting evidence for the exclusion restriction underlying the instrument. If exposure to non-core lenders operates through credit supply, counties with different levels of $NCL_{c,2002}$ should exhibit similar trends in housing market features, including product type, prior to the 2003 interest rate shock. Existing evidence in [Mian and Sufi \(2021\)](#) supports this interpretation. Using fine-grained spatial fixed effects, they show that areas with greater concentrations of NCL lenders do not experience differential demand shocks around the initial interest rate increase.

3.B PLS Market Event Study Design Results

Our IV regressions, shown in Table 1, show that higher exposure to NCL lenders prior to the housing boom is a powerful instrument for changing the product composition of mortgages and increasing payment affordability in a county.¹⁷ The effect on changing product composition is shown through comparing the first stage F-statistics for each measure of credit supply. The highest first-stage F-statistic for any variable we study is the change in the percent of mortgages originated as interest-only. NCL exposure is also a strong instrument for negatively amortizing loans. Correspondingly for payments, the results show that NCL exposure strongly predicts the size of the median payment gap and 12-month payment appreciation. Beyond product types, NCL exposure is a strong instrument for the prevalence of second liens and the transition from FHA to subprime lending. Of much weaker relevance are the extension of teaser loans or the deterioration of other common mortgage attributes of owner occupancy, documentation, LTV, and FICO scores.

¹⁷Appendix Table A2 presents the OLS regression outcomes for comparison. The IV estimates reveal a stronger relationship between non-traditional mortgage products and house prices compared to the OLS estimates. This difference is consistent with the presence of omitted variable bias. Specifically, omitted variables that are positively correlated with both credit supply and house prices, such as increased investor activity or changing expectations for house price growth, can negatively bias the OLS results.

Table 1: IV Estimates for Changes in House Price Appreciation and Credit Supply

Panel A: Alternative Financing Features

	Balloon (1)	Interest Only (2)	Extended Term (3)	Neg. Amortization (4)	Teaser (5)	Traditional ARM (6)
HPA Percent Change 2002m1 to 2005m4	-25.919 (233.064)	0.195*** (0.015)	1.376*** (0.522)	0.266*** (0.023)	0.697*** (0.196)	-1.165** (0.477)
First stage F-stat	0.01	167.03	6.37	87.32	12.43	6.19
Mean Change	-1.026	14.363	0.791	3.244	15.494	0.085
Std Deviation Change	2.576	14.191	3.297	5.503	10.536	6.916
N	1156	1156	1156	1156	1156	1156
R-Squared	.	0.244	.	0.007	.	.

Panel B: Payments

	Pmt. Gap (1)	Pmt. Gap, Any (2)	Med. Interest Rate (3)	12-Month Pmt. Appr. (4)
HPA Percent Change 2002m1 to 2005m4	-0.407*** (0.057)	-0.442*** (0.072)	-2.303 (1.837)	0.297*** (0.025)
First stage F-stat	51.83	41.00	1.53	100.18
Mean Change	11.646	-38.849	-0.907	4.696
Std Deviation Change	115.198	240.641	0.319	7.669
N	1150	879	1155	1162
R-Squared	.	.	.	0.009

Panel C: Other Mortgage Attributes

	Owner Occupied (1)	Full Documentation (2)	LTV (3)	FICO (4)
HPA Percent Change 2002m1 to 2005m4	-0.555*** (0.109)	-0.767*** (0.230)	-0.583*** (0.132)	1.061** (0.420)
First stage F-stat	25.78	10.90	20.13	6.38
Mean Change	-4.295	-0.670	0.930	0.807
Std Deviation Change	8.325	26.219	5.430	26.210
N	1162	1097	1156	1151
R-Squared	.	.	.	.

Panel C (Cont.): Other Mortgage Attributes

	FHA (1)	Subprime (2)	Second Lien (3)	Denied (4)
HPA Percent Change 2002m1 to 2005m4	-0.354*** (0.050)	0.299*** (0.039)	0.230*** (0.025)	0.441*** (0.069)
First stage F-stat	60.08	60.52	129.32	42.20
Mean Change	-6.223	4.957	5.029	1.816
Std Deviation Change	7.719	5.624	6.610	6.307
N	1162	1162	1162	1162
R-Squared	.	.	.	.

Note: Significance levels denoted by * 0.10 ** 0.05 *** 0.01. This table presents the results of equation 4. House price appreciation and each measure of credit supply are standardized by their mean and standard deviation, shown as a summary statistic for each regression. First stage F-statistics for the each measure of credit supply in a county instrumented with county NCL exposure are also displayed. Negative R-Squared statistics are suppressed. OLS regression results available in Table A2.

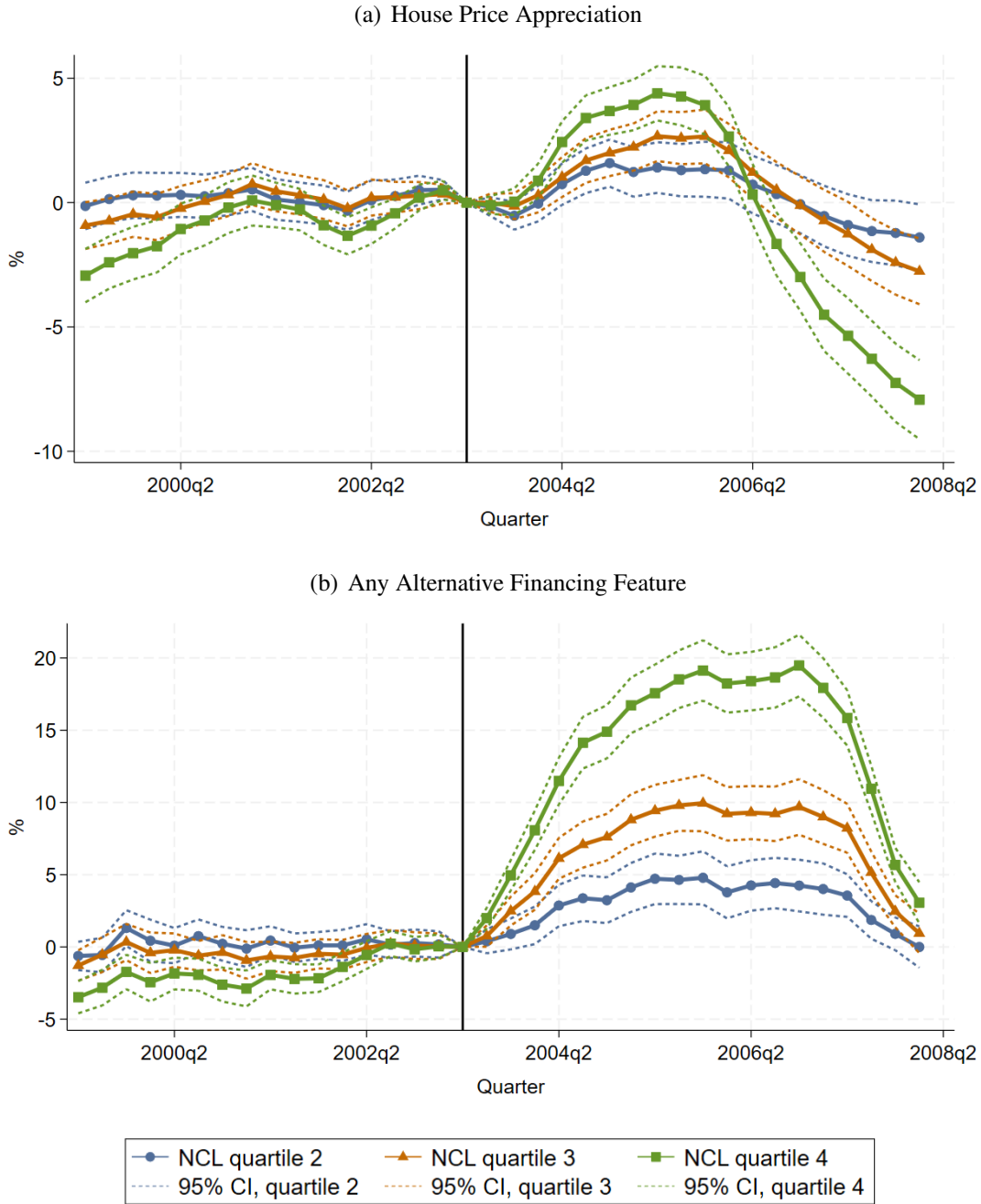
Figures 4 - 7 show the results of our equation 5 difference-in-differences specification for house prices, alternative features, payment features, and other mortgage attributes. Through each figure we see that prior to the interest rate shock, counties across NCL exposure quartiles show largely parallel trends. This robust pattern suggests that counties with differing NCL exposure had otherwise similar housing market conditions, including in the types of mortgage products originated, and that the 2003 rate increase was plausibly exogenous to those conditions.¹⁸

After the 2003 Treasury rate shock, we observe a distinct divergence in house price appreciation and mortgage financing across NCL exposure quartiles. Figure 4 panel (a) shows that counties in the highest quartile of NCL exposure experience house price appreciation rates over 4 percentage points higher than those in the lowest quartile at the boom's peak. However, this difference does not meaningfully emerge until three quarters after the shock. Conversely, panel (b) reveals a sharp, immediate uptick in the use of non-traditional mortgage products across quartiles, with more pronounced increases occurring in counties more exposed to NCL. Specifically, the gap in non-traditional product use between the highest and lowest quartiles expands from 2 percentage points in the first quarter after the shock to 11.5 percentage points by the fourth quarter. A closer examination in Figure 5 reveals that this surge is primarily fueled by interest-only loans and, to a lesser extent, teaser-rate ARMs and loans allowing for negative amortization. For comparison, traditional ARMs show no differential increase, suggesting that they were entirely supplanted by non-traditional mortgages in this period of rising interest rates. Furthermore, the timing of these shifts, with non-traditional mortgage use increasing before house price appreciation, strongly indicates that alternative financing mechanisms were instrumental in the post-2003 surge in house prices.

Figure 6 provides evidence that non-traditional mortgage products and traditional ARMs dramatically reduced mortgage payments at origination. Panel (a) illustrates the average payment gap across different levels of NCL exposure. From the onset of the interest rate shock to the peak of the housing boom, counties with the highest NCL exposure saw monthly mortgage payments become, on average, \$72 more affordable compared to traditional 30-year FRMs, relative to the least exposed counties. This affordability was not just due to the widespread selection of non-traditional products over traditional ones, but also because non-traditional contracts themselves became cheaper. As detailed in Panel (b), upon securing a non-traditional loan, borrowers in the most NCL-exposed counties saved an additional average \$61 on their mortgage payments after the interest rate shock up until the boom's peak.

¹⁸Denial rates and the percent of mortgage originated as subprime show small differences in pre-trends prior to 2000. When we add these outcomes as controls for the alternative financing features, there are no differences in our main results.

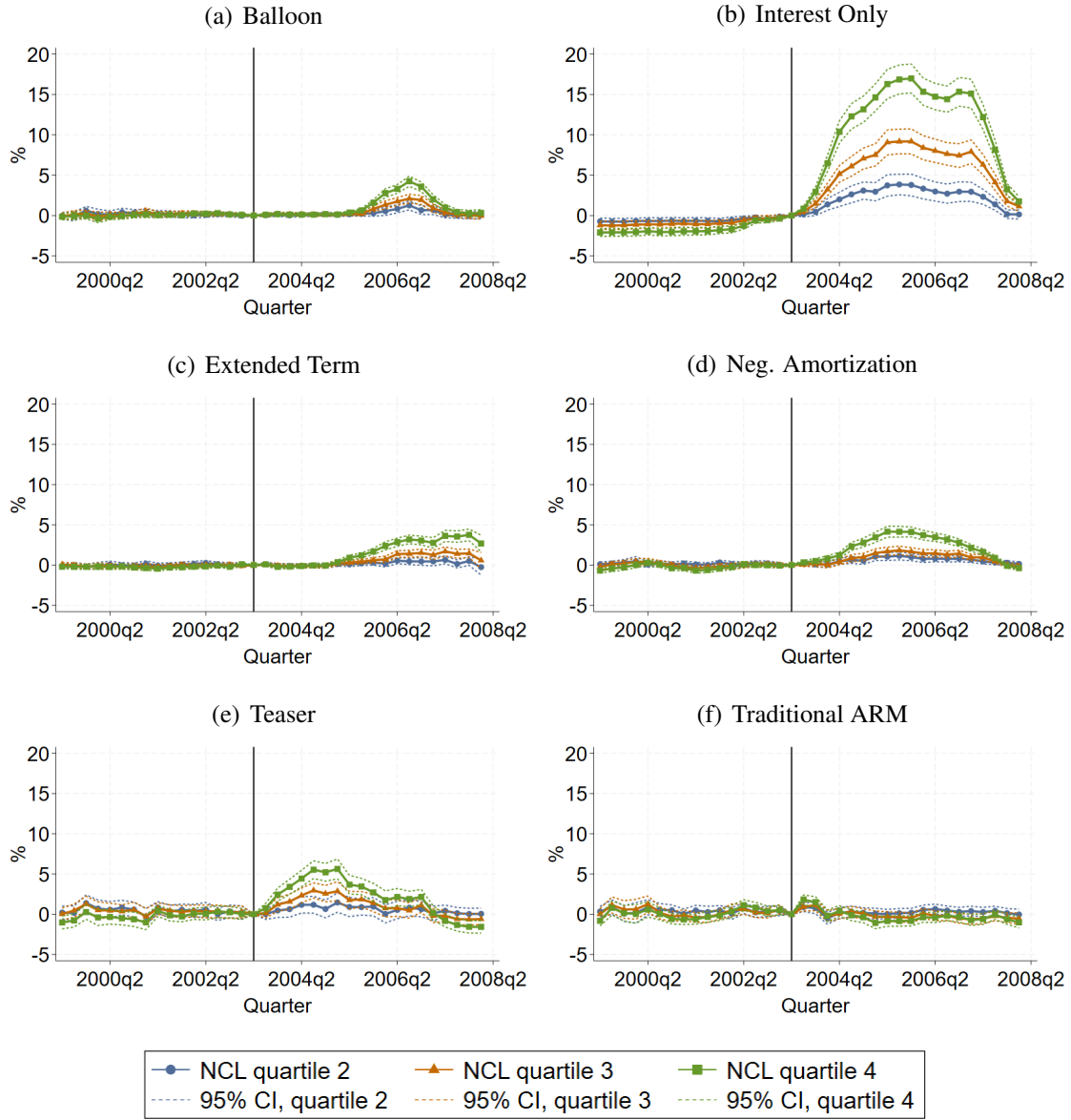
Figure 4: House Price Appreciation and Alternative Financing Features by Non-core Lending Quartile



Note: This figure shows house price appreciation and the use of alternative mortgage products around the mid-2003 acceleration of the private label securities market. Counties are divided into quartiles of exposure to non-core liability lenders. The coefficients are derived via equation 5 and are relative to the first (bottom) quartile of counties.

Further insights into mortgage payments are offered in Panels (c) and (d). Panel (c) shows that in counties with the highest NCL exposure, average median interest rates at origination fell

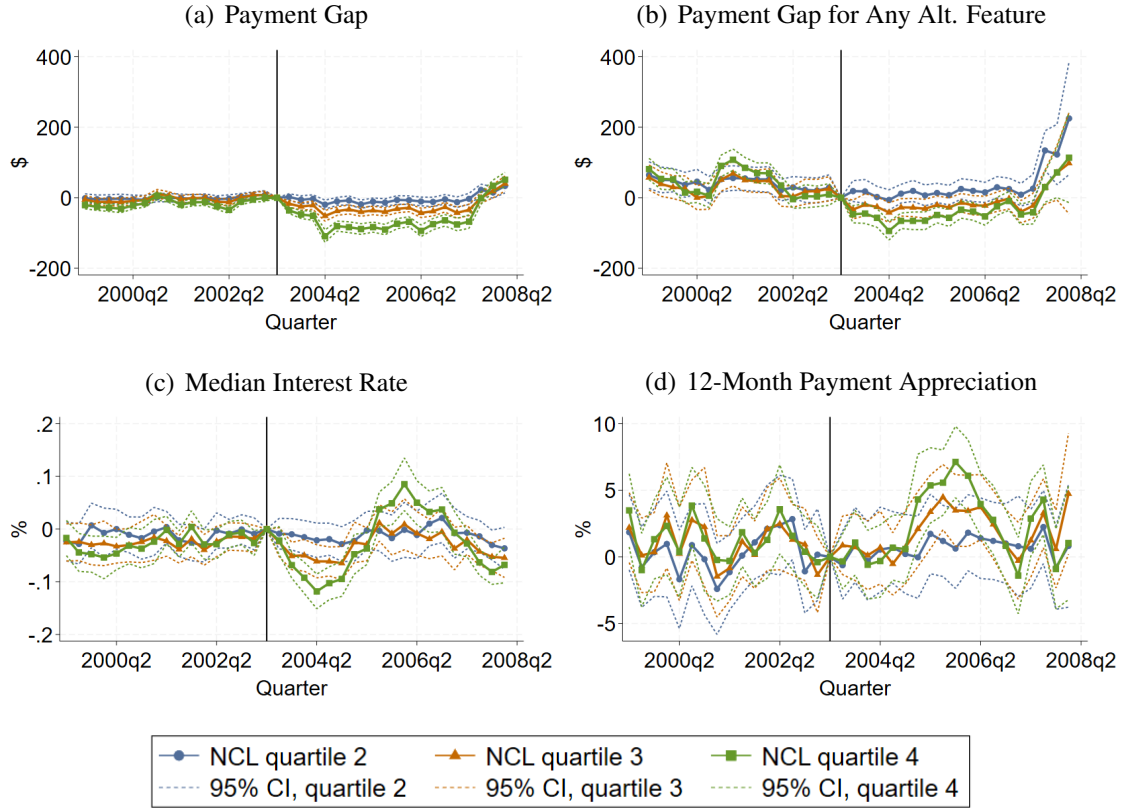
Figure 5: Alternative Financing Features by Non-core Lending Quartile



Note: This figure shows the share of purchase loans originated with each alternative financing feature around the mid-2003 acceleration of the private label securities market. Traditional ARMs shown for comparison. Counties are divided into quartiles of exposure to non-core liability lenders. The coefficients are derived via equation 5 and are relative to the first (bottom) quartile of counties.

relative to counties with the least exposure for two years. The difference was most pronounced in the second quarter of 2004, with a more than 10 basis-point lower average median interest rate in these counties. Panel (d) addresses how mortgage product composition changes affected mortgage payment appreciation across counties with varying NCL exposures. Through changes in product composition, there was no differential change in mortgage payment appreciation across

Figure 6: Payments by Non-core Lending Quartile



Note: This figure describes changes in payments around the mid-2003 acceleration of the private label securities market. Counties are divided into quartiles of exposure to non-core liability lenders. The coefficients are derived via equation 5 and are relative to the first (bottom) quartile of counties.

counties until both house price appreciation and median interest rates also differentially increased. The timing of these increases in mid-2005 coincides with the differential increase in balloon and extended-term mortgages observed in Figure 5. This coincidence suggests that these features, often paired with interest-modifying elements, were used as secondary measures for maintaining payment affordability when interest-modifying features alone could not curb payment appreciation.

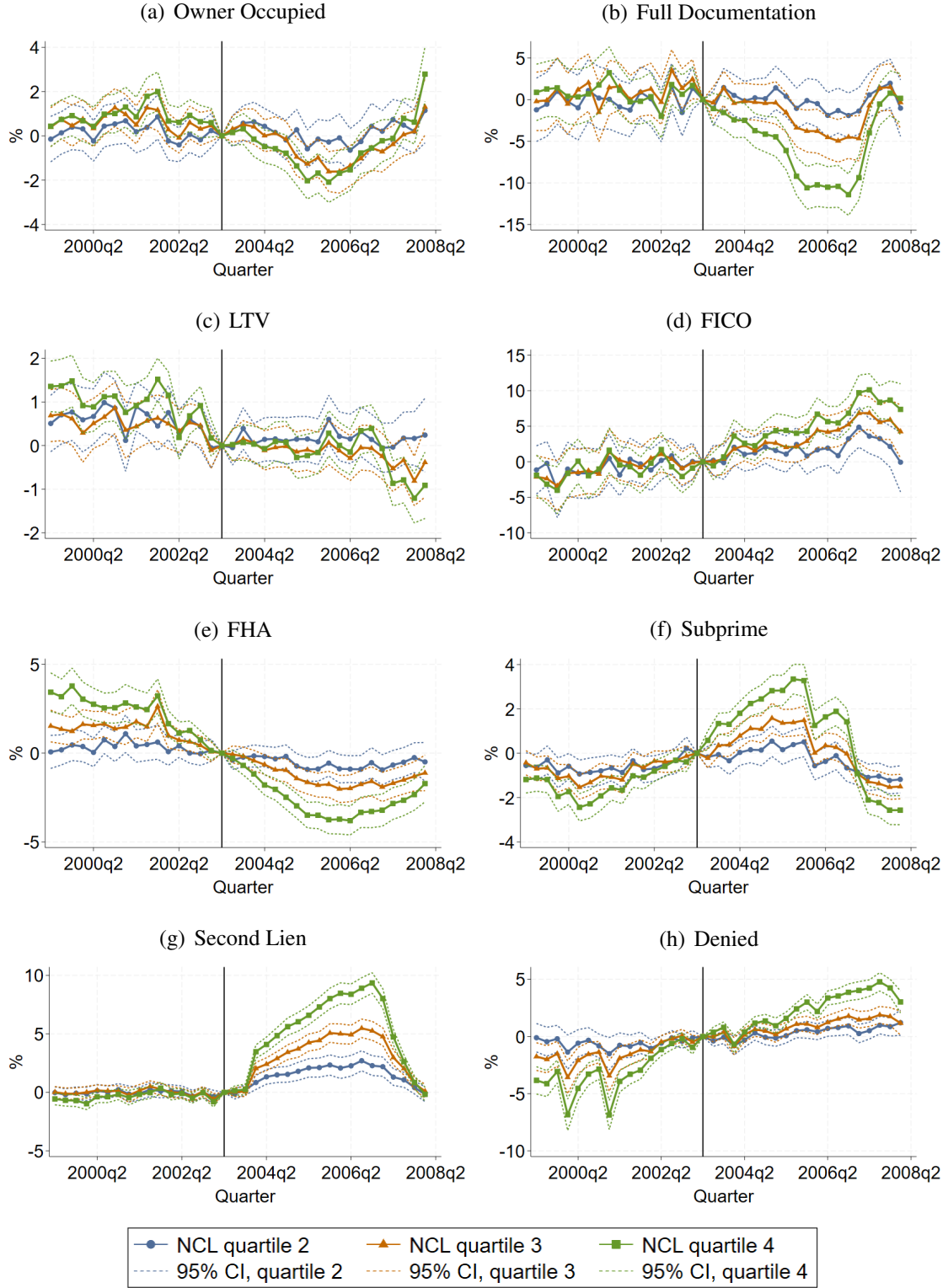
Figure 7 shows evidence of differential expansion in the credit supply in more NCL-exposed counties along other margins that have been the focus of the literature. The more exposed a county to NCL, the more substantial the decline in owner occupancy, full documentation, and FHA market share, and the more substantial the increase in seconds liens and subprime loans. Tellingly, there is little appreciable difference in LTV ratios or FICO scores, margins which would have been more readily apparent to mortgage-backed securities (MBS) investors. If anything, more NCL-exposed counties show an increase in average FICO score at origination, consistent with a shift in composition towards high-credit quality non-owners (Mian and Sufi 2021). Finally, we observe that through the decade, denial rates increased faster in more NCL-exposed counties, suggesting

that demand continued to outstrip supply despite the supply expansion after the Treasury rate shock.¹⁹

In sum, we find that counties with higher exposure to the PLS market experienced a sharper rise in the use of non-traditional mortgage products immediately after the 2003 rate increase, keeping payment appreciation in these markets at parity for markets with lower exposure. These shifts crucially preceded any divergence in house prices across more PLS-exposed counties and have the strongest association with that exposure. At the same time, lenders in more exposed markets changed their credit supply along other dimensions that were also opaque to MBS investors and shifted from FHA to subprime mortgages. This evidence suggests that non-traditional product use was an important mechanism relative to alternative explanations in transmitting the credit supply expansion from the PLS market to housing markets.

¹⁹Figure A5 indicates no significant differences in income or unemployment improvements in counties more exposed to NCL immediately following the 2003q2 interest rate shock. The delay in economic improvement until after house prices differentially accelerated implies the housing market might have spurred broader economic gains, rather than the converse. Justifying the increase in credit and house prices in 2003 would have required borrowers and lenders to correctly forecast these improvements, though sharp changes in interest rates would normally generate anticipation of economic decline.

Figure 7: Other Mortgage Attributes by Non-core Lending Quartile



Note: This figure shows patterns in other mortgage attributes around the mid-2003 acceleration of the private label securities market. Counties are divided into quartiles of exposure to non-core liability lenders. The coefficients are derived via equation 5 and are relative to the first (bottom) quartile of counties.

4 Evidence from Early vs. Late Housing Booms

In this section, we study whether the rising use of non-traditional mortgage products more systematically precedes rising house prices outside of the narrow shock around 2003. As Figure 1 shows, while house price appreciation and the use of non-traditional mortgage products peak at roughly the same time during the housing boom, non-traditional products were originated since the early 1990s. This aggregate relationship presents an opportunity to understand whether non-traditional products also contributed to early booms in a way that has gone unappreciated until now or if the connection is special to the mid-2000s. We do so by identifying the start of local housing booms and studying the changes in non-traditional product use and non-product attributes of mortgage supply for early- versus late-booming markets.

4.A Identifying Housing Booms

Our approach is to identify the start of a county's housing boom by using a breakpoint randomization methodology that selects the month in which the 12-month house price appreciation rate first deviates most sharply from prior trends. We interpret this large deviation from trend as indicating that the forces driving local house prices changed in a meaningful way. This follows the approach taken by Hansen (2000) and applied to metropolitan housing markets by Charles, Hurst and Notowidigdo (2018), DeFusco et al. (2018), and Ferreira and Gyourko (2023).

For each county c and month m , we run regressions over all possible breakpoints λ_{cm}^* of the form:

$$\Delta P_{cm} = \alpha_c + \beta_c \mathbf{1}[\lambda_{cm} \geq \lambda_{cm}^*] + \varepsilon_{cm} \quad (6)$$

In the sample for each county, we only include the months up to and including the month with the largest 12-month increase in house price appreciation. This restriction ensures our methodology chooses a structural break associated with the boom in house prices, rather than a break associated with the later bust. Practically, this methodology usually involves running somewhere between 60 and 150 regressions (one for each month observed for a given county from the beginning of the data to the house price appreciation peak) for each of the 1,163 counties with an available house price index from CL. From each regression, we recover the sum of squared errors, $S(\lambda_{cm})$ and identify the structural breakpoint in house price growth as:

$$\lambda_{cm}^{**} = \operatorname{argmin}(S(\lambda_{cm})) \quad (7)$$

We exclude any county for which this chosen breakpoint is not significant at the 5 percent level.²⁰ This leaves us with 984 counties in our main analysis sample. Further details on the breakpoint methodology are provided in Appendix Section 6.D.

We separately analyze counties with housing booms that started before and after February 2002 when the national house appreciation series in Figure 1 reached its pre-boom nadir. This cutoff is motivated both by the structural changes in the PLS market in 2003 described above, and the fact that early- and late-booming counties are concentrated in fundamentally different geographic areas. Markets with booms before February 2002 occurred in more inelastic, urban, and coastal markets, such as San Francisco and Boston, and had a history of previous booms (DeFusco et al. 2018; Ferreira and Gyourko 2023). In contrast, later booming markets, like Las Vegas and Phoenix, had a relatively elastic housing supply and little to no history of extreme housing cycles since World War II (Himmelberg, Mayer and Sinai 2005; Sinai 2013). Among 984 counties with statistically significant house price breaks, 613 experienced early booms, and 371 saw late booms.

Figure 8 demonstrates that our methodology successfully identifies the start of housing booms. Early-booming markets experience an average increase in appreciation of 5.8 percentage points, relative to 5.5 percent in the year before. Late-booming markets experience an average increase of 8.8 percentage points, relative to 8.0 percent in the year before. These results confirm the sharp discontinuities our method detects, with minimal differences in house price appreciation preceding the boom.

4.B Housing Boom Event Study Design

After identifying the start of each county’s house price boom, we conduct an event study analysis of changes in alternative features, payment features and additional credit attributes:

$$Y_{cm} = \alpha_c + \beta f(\lambda_q^*) + g(m) + \gamma X_{cm} + \epsilon_{cm} \quad (8)$$

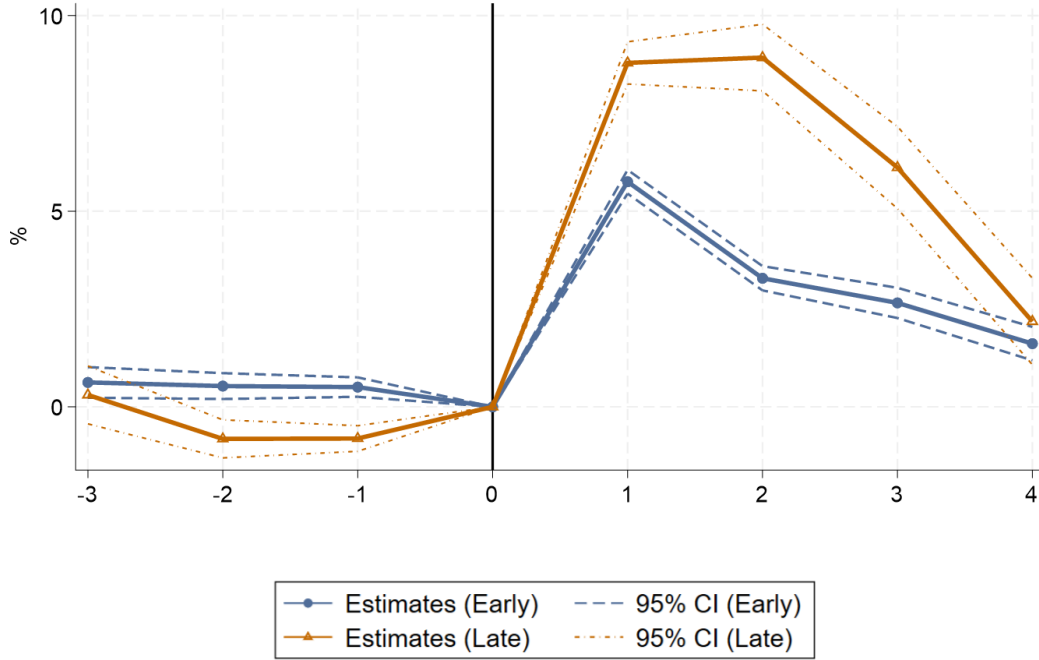
where Y_{cm} is an outcome in county c and month m , α_c is a county fixed effect, $g(m)$ is a set of calendar time controls, X_{cm} is a vector of county economic fundamentals (income level and unemployment rate), and $f(\lambda_q^*)$ is a vector of relative-quarter dummies.²¹ We normalize the dummies so that the quarter of the house price break in county c is the omitted category (and the corresponding coefficient set to 0). We estimate equation 8 separately for early- and late-breaking counties.²²

²⁰In robustness analysis, we have made our significance threshold more restrictive and our conclusions are little changed. These results are available upon request.

²¹In specifications where the dependent variable is an economic fundamental, we omit the additional X_{cm} controls.

²²Since household income is only collected annually, we linearly interpolate monthly values between years when used as an outcome variable in the event study. We do not control for income and unemployment for these as outcome variables.

Figure 8: House Price Appreciation around House Price Breaks

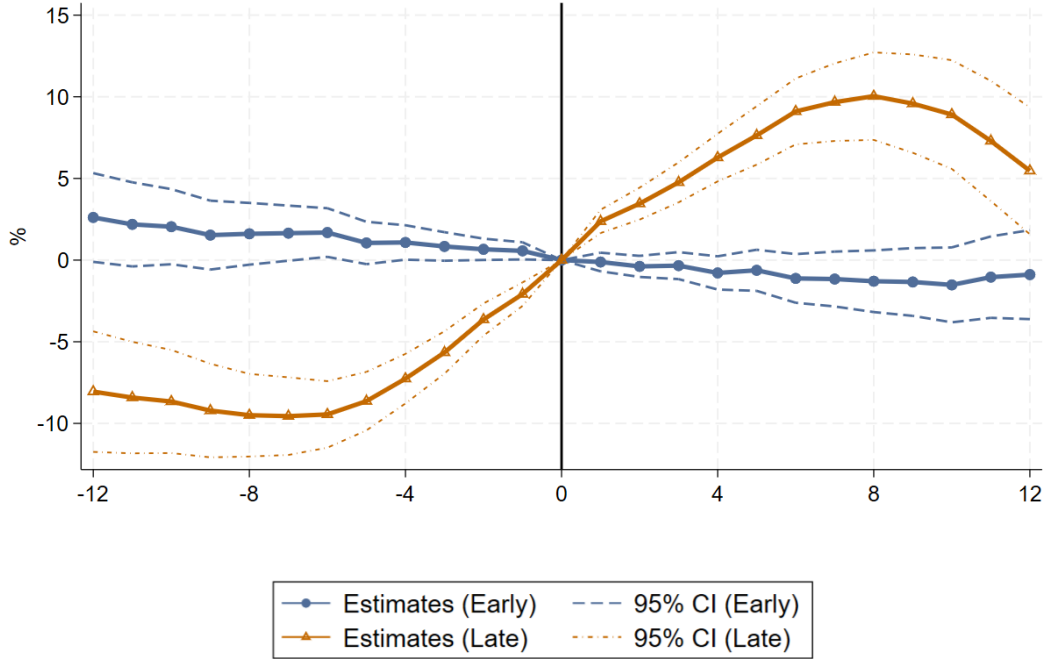


Note: This figure plots the β coefficients from the specification $HPA_{cm} = \alpha_c + \beta f(\lambda_y^*) + g(m) + \gamma X_{cm} + \varepsilon_{cm}$, where HPA_{cm} is 12-month house price appreciation, α_c are county fixed effects, $f(\lambda_y^*)$ are dummies relative to the year of the house price boom, $g(m)$ is a set of calendar time controls, X_{cm} are vectors for income and unemployment. The year at the start of the housing boom, $y = 0$, starts in the month of the house price breakpoint. Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals. We show a yearly event study for house price appreciation because the start of local housing booms are defined using year-over-year appreciation.

The inclusion of county fixed effects, local fundamentals, and time trends account for important covariates that could impact both non-traditional mortgage use and house prices. For example, [Amromin et al. \(2018\)](#) find that borrowers using interest-only and negative-amortization mortgages tend to have higher income and more education. Furthermore, because households geographically segregate along these dimensions that also matter for housing markets, it is important to estimate the relationship between house prices and mortgage features within counties over time. Controlling for calendar time accounts for underlying national trends in mortgage and borrower characteristics so that the estimated relationship between house price breaks and these attributes is independent of their macroeconomic trends. Finally, including shocks to local economic fundamentals, specifically income and unemployment, accounts for time-varying local conditions. Therefore, the β coefficients tell us, within county and apart from pre-existing national trends and local changes in economic fundamentals, the difference in appreciation, mortgage attributes, or credit supply in the quarters around the start of the local house price boom.

4.C Results

Figure 9: Non-Traditional Mortgages around House Price Breaks

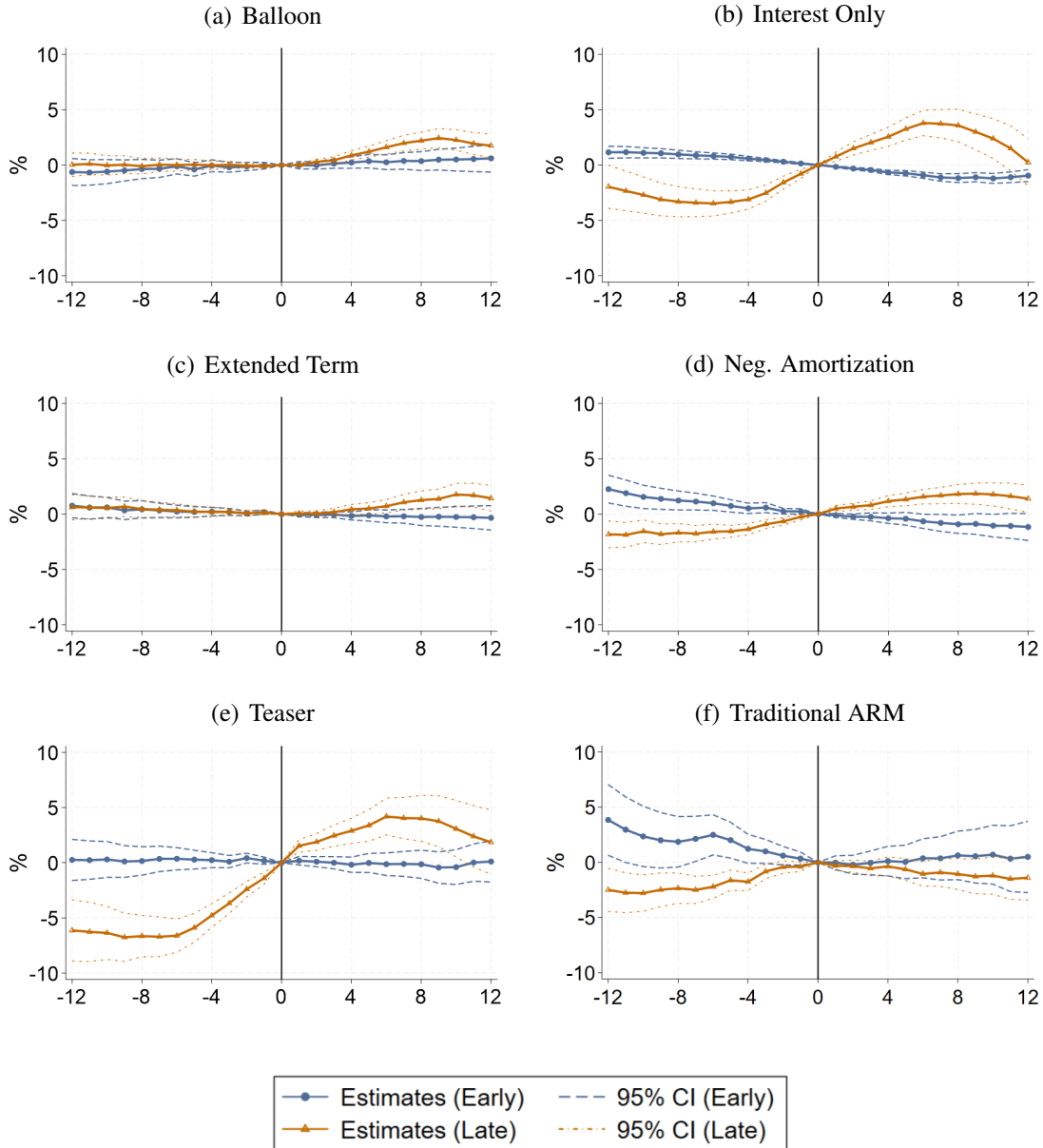


Note: This figure shows the estimation results for equation 8 for the share of loans with any alternative financing feature. The start of the the housing boom is $t = 0$ (defined as the quarter of the structural break in house prices). Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals.

Using this event study design, we first investigate non-traditional mortgage product use around the start of early- and late-booms across counties. The results are shown in Figures 9 and 10. We find that early-booming markets show no notable shift toward the use of non-traditional mortgage products before the start of booms. If anything, there is a modest decrease in their adoption in the periods leading up to and through these early booms. The most notable structural break coincident with the start of early booms is the end of the steady pre-boom decline in the share of traditional ARMs. This suggests that traditional ARMs were more attractive to borrowers to moderate mortgage payments. Indeed, in Appendix Figure A15, we find that this adoption of traditional ARMs once prices began to boom in early periods is concentrated outside subprime borrowers for whom traditional lenders may have been more willing to make more affordable, though riskier, loans.

In contrast, we find that increased use of non-traditional products strongly predates rapidly rising house prices in late-booming markets. In late-booming markets, extensive use of non-traditional mortgage products shows a sizable and quantitatively significant increase of 7.3 percentage points in the four quarters preceding the start of the booms and for two years thereafter.

Figure 10: Alternative Financing Features around House Price Breaks

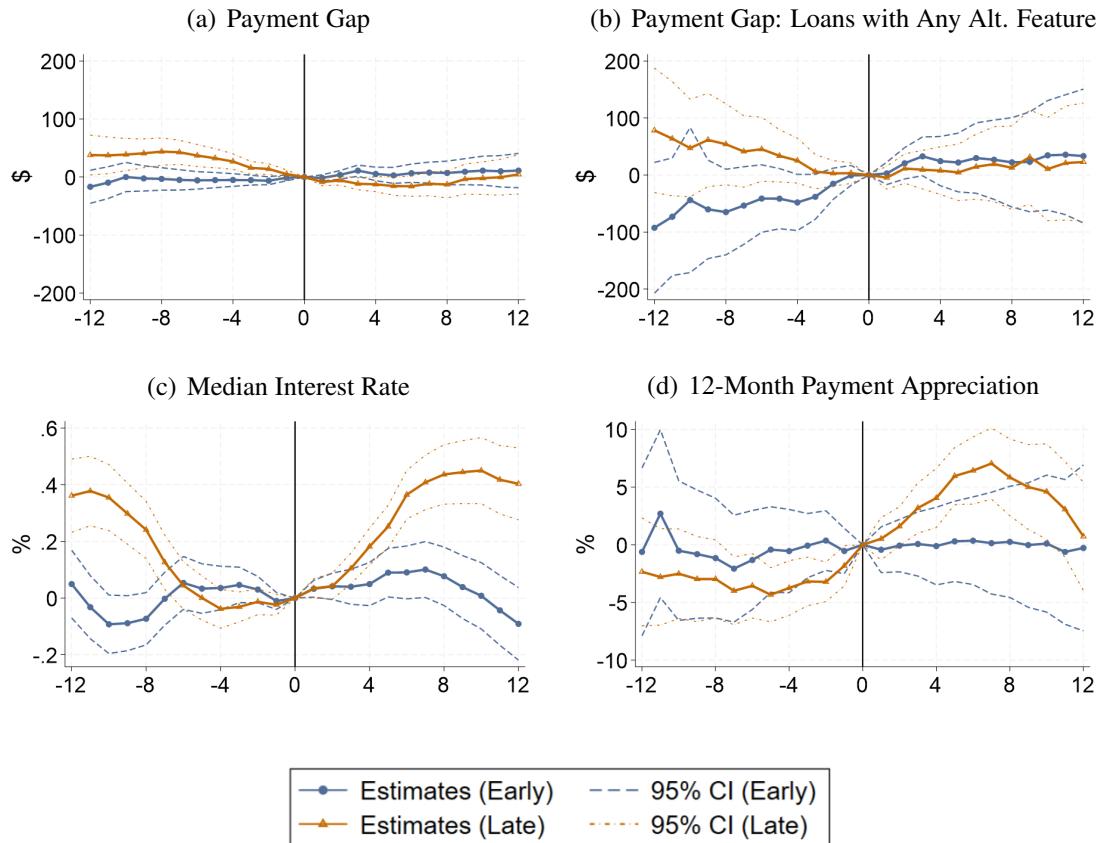


Note: This figure presents event study results for alternative financing features. The start of the the housing boom is $t = 0$ (defined as the quarter of the structural break in house prices). Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals. We show a yearly event student for house price appreciation because the start of local housing booms are defined using year-of-year appreciation.

Inspecting our results for specific non-traditional products, we again find that interest-only, teaser, and negative-amortization loans drive the shift toward non-traditional mortgage products prior to the start of late booms. These features increase by 3.1, 4.8, and 1.4 percentage points, respectively, in the four quarters before the boom. Mortgages with balloon payments and extended terms

only increase in popularity after the start of most house price booms. We also observe a small statistically significant increase in traditional ARMs ahead of the start of these booms similar in magnitude to that for negatively amortizing loans. However, unlike negatively amortizing loans, traditional ARMs begin to decline in use during the boom. This pattern further suggests displacement of traditional ARMs in favor of alternative products that brought larger payment savings.

Figure 11: Payments around House Price Breaks



Note: This figure presents event study results for payment features. The start of the the housing boom is $t = 0$ (defined as the quarter of the structural break in house prices). Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals. We show a yearly event student for house price appreciation because the start of local housing booms are defined using year-of-year appreciation.

Figure 11 shows the evolution of mortgage payments at origination. In early booming markets, we observe relative stability in average payment gaps, median interest rates, and payment appreciation around the boom's onset. Interestingly, the payment gap for borrowers opting for mortgages with alternative financing features fell to zero in the years before the start of early housing booms and these mortgages offered no reduced payments during those booms. Conversely, in markets that boomed later, payment trends exhibit a clear alignment with the aftermath of the 2003 Treasury rate

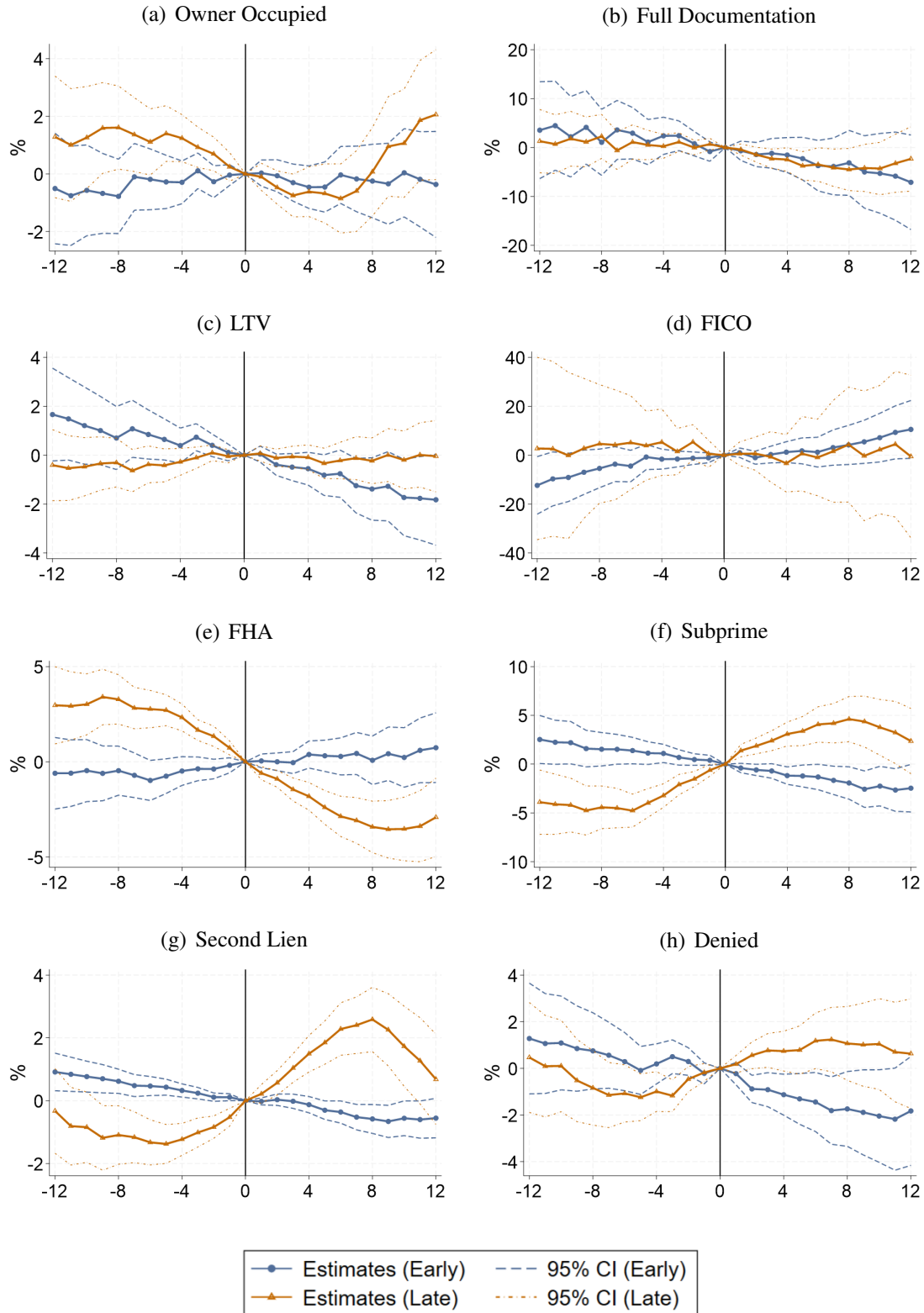
adjustments. Significantly, we note a decrease in average median interest rates of approximately 24 basis points in the two years before the start of the late booms. This decline coincides with a decline in the gap between actual mortgage payments and their counterfactuals by an average of \$44. Not until house prices began their acceleration did average contracted payment growth started to accelerate, at a rate only half that of house price acceleration.

We find that early- and late-booming markets also show important differences along non-product dimensions of credit supply. These results are presented in Figure 12. For early-booming markets, we find that underwriting standards moderately tighten around the start of local housing booms across each of our measures. For instance, LTV ratios, second liens, and the market share of subprime versus FHA loans decline steadily. These trends are all smoothly linear in the years before and after, suggesting little association with the timing of the start of the boom. In stark contrast, late-booming markets show non-linear trends in underwriting, suggesting deteriorating quality. Consistent with our analysis around the 2003 Treasury rate increase, in the four quarters before the starts of late booms, there are substantial declines in owner-occupancy and FHA market share, while the subprime share and second lien share both increase.²³

Finally, we conduct several robustness checks of the early and late housing boom event study results. First, we examine changes in local economic fundamentals around the start of housing booms using a version of equation 8 that excludes income and unemployment as controls. Second, we test the sensitivity of our results to allowing counties to experience multiple housing boom breakpoints. Third, we examine whether changes in non-traditional mortgage use differ across borrower credit quality by estimating our event study separately by FICO score groups. The results of these exercises are presented in Appendix Section 6.E.

²³Figure A10 shows little evidence of improving economic fundamentals before either early or late housing booms. After the start of the booms, fundamentals deteriorate in early-booming markets but improve in late-booming ones.

Figure 12: Other Mortgage Attributes around House Price Breaks



Note: This figure presents event study results for other mortgage attributes. The start of the the housing boom is $t = 0$ (defined as the quarter of the structural break in house prices). Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals. We show a yearly event student for house price appreciation because the start of local housing booms are defined using year-of-year appreciation.

5 Conclusion

During the housing boom of the 2000s, the growth in the use of non-traditional mortgage products was as spectacular as the rise in house prices. In this paper, we examine the role of non-traditional mortgages as a mechanism that facilitated the credit expansion that fueled that rise. To do so, we compile a novel historical dataset on county-level house prices and mortgage contracts, and use transparent methods to study the relationship between non-traditional products and house price accelerations. We also construct a novel measure of payment affordability to highlight the impact of non-traditional mortgage products in expanding credit supply through their reduction in monthly mortgage payments. This method for measuring payment affordability via the payment gap can be used in ongoing evaluations of mortgage credit. The data requirements consist only of basic borrower and loan characteristics and a representative sample for the estimation of a counterfactual payment in a base period for which changes in affordability can be measured against.

In our empirical work, we take two approaches that allow us to study the the relationship between alternative products and house prices. We first exploit the quasi-experimental differential impact of rising Treasury rates in 2003 on areas with varying degrees of securitization activity to show that non-traditional mortgage use preceded the rise in house prices and correlated with the intensity of ex-ante NCL exposure. Our broader event study analysis of markets in early-versus late-booming counties then studies whether this relationship is systematic across time and geographic markets. Only late-booming markets show correlations in alternative product use with the start of local housing booms, suggesting that the financial liberalization of the 2000s was an important contributing factor.

Thus, our research uncovers a fundamental “product channel” through which financial liberalization affected house prices by moderating payment appreciation. We also find these patterns informative of lenders’ strategic decisions in loan originations for the PLS market in this period. First-lien LTV and FICO scores constitute hard information that would be reported to MBS investors, while only the lender would observe more soft information about the borrower ([Rajan, Seru and Vig 2015](#)). Consistent with the results of [Keys et al. \(2010\)](#); [Rajan, Seru and Vig \(2015\)](#), despite a shift toward loans originated by subprime lenders, these hard information measures show little sign of deterioration around booms. Even after house price accelerations, most of these hard information measures are unchanged. However, we document that underreported dimensions do indicate quality declines, such as the increased use of second liens ([Piskorski, Seru and Witkin 2015](#); [Griffin and Maturana 2016](#); [Ghent, Miltersen and Torous 2020](#)). Detailed information on product composition, while technically reportable, may have been unfamiliar or obscure if only

basic aggregate information was closely monitored by investors. This opacity may have incentivized lenders to innovate along the product dimension in their credit supply.

In this work, we focus on periods prior to accelerating house prices. However, our approach intentionally omits the feedback effects these products may have created during the bubble period, in which additional waves of borrowers may have chosen non-traditional products to maintain affordability in markets with already accelerating house prices. Disentangling the unique contribution of product choice from other dimensions of credit supply, especially in a dynamic context, also presents challenges. Whether there were important feedback effects between product mix, other dimensions of credit supply, and prices remains a topic for future research.

In the wake of the crisis, regulators placed significant limitations on a number of the alternative mortgage features discussed in this paper by excluding them from the definition of a Qualified Mortgage (QM) for regulatory purposes. As laid out by the Consumer Financial Protection Bureau (CFPB) in their January 2013 report, lenders are no longer able to underwrite a qualified mortgage loan based on a teaser rate in determining ability to pay.²⁴ Moreover, mortgages cannot contain interest-only, negative-amortization, or extended term features if lenders want to meet the legal presumption of complying with the qualified mortgage regulation.

As we have shown, whether restricting these contract features will affect the formation and magnitude of housing price cycles in the future depends crucially on the context of the use of these features. Indeed, for early booms before the rise of the PLS market, we find no evidence that these products spurred housing booms independently. Furthermore, limiting these products is likely to reduce credit access for some households, while guiding others into more standard contracts that may be suboptimal. Understanding how incentives shape lending decisions and hardening the information pipeline between lenders and investors may ultimately be a more robust policy response. We propose monitoring the payment gap as another policy tool. These policies can complement other efforts to encourage mandatory mortgage counseling, so that both borrowers and lenders better understand their risk profile (Agarwal et al. 2014). The consequences of these regulatory changes for the interaction between non-traditional mortgage products and house prices is a promising area of future research.

Finally, current housing market conditions show that banning these products does not eliminate housing booms altogether. In this environment, in which both prices and interest rates are rising, lenders and borrowers will face pressure to find new ways to maintain payment affordability. This pressure may result in the resurgence of a private securities market which supports more affordable products or growth in other non-QM markets that could increase systemic risk in the housing market.

²⁴See CFPB (2013) on ability to repay rules and qualified mortgages.

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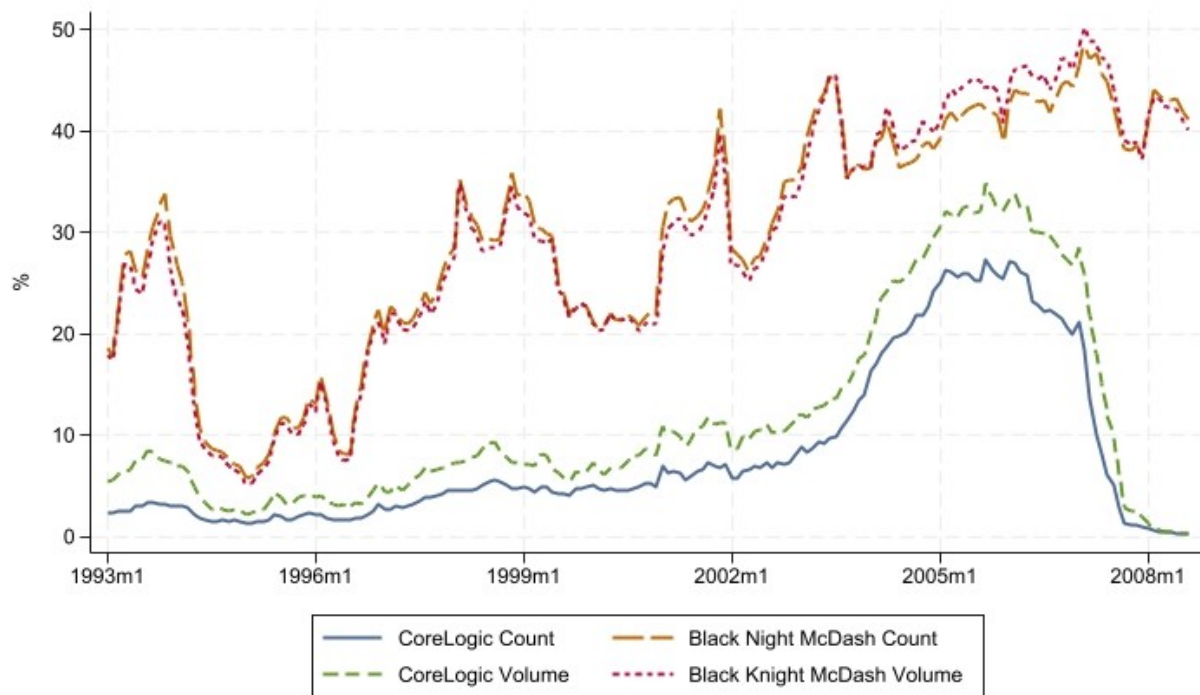
6 Supplemental Appendix

6.A Data

This subsection provides additional detail on the coverage of the loan-level data and summary trends in key mortgage attributes. Figure A1 compares the coverage of the CoreLogic and Black Knight McDash servicer data to HMDA over time, showing both loan counts and origination volume. As discussed in Section 2, coverage of the servicer datasets increases substantially during the late 1990s and early 2000s as additional servicers are incorporated. Figure A3 then presents national time-series trends for other mortgage attributes that we use to characterize credit supply conditions in our analysis. These figures provide additional context for the evolution of mortgage market composition during the housing boom.

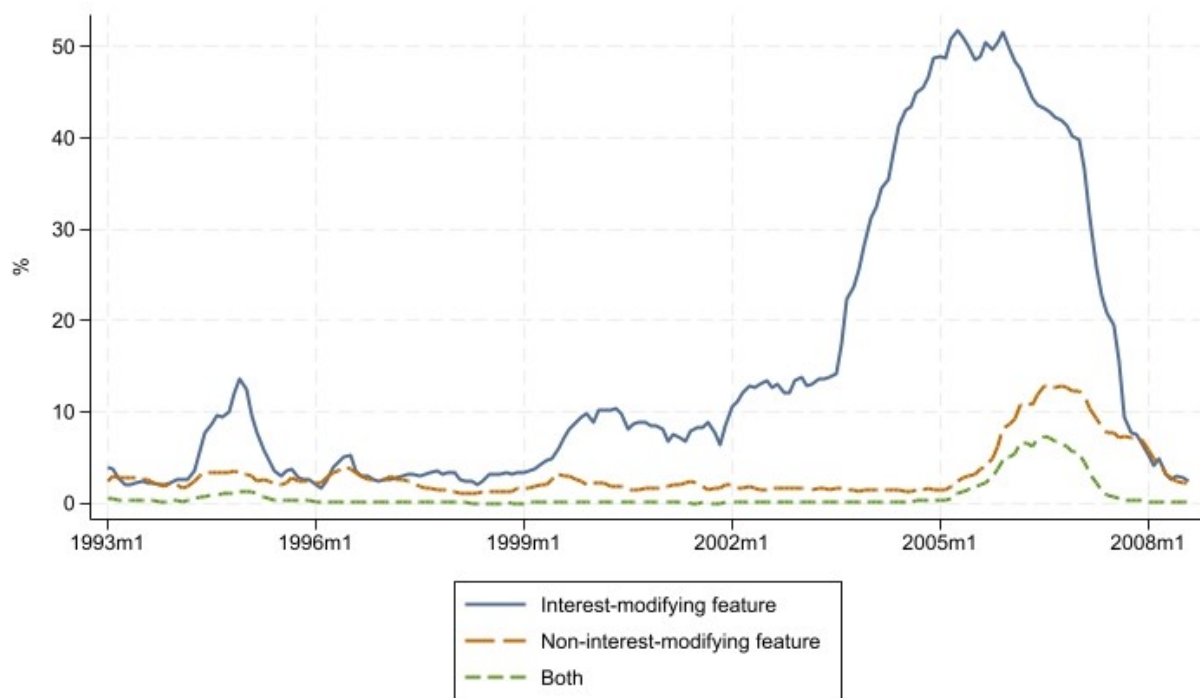
Subprime loans are identified by matching HMDA agency and lender codes with the subprime lender list created by HUD. According to Mayer and Pence (2009), this approach provides a reasonable measure of the size of the subprime market. Although HMDA includes an indicator for “high-cost” loans in later years, this measure does not necessarily correspond to subprime status because HMDA does not directly observe borrower creditworthiness. FHA loans differ from subprime loans in several important ways. FHA-insured mortgages are typically 30-year fixed-rate loans that require a minimum downpayment of 3.5 percent and are subject to limits tied to the conforming loan limit. Subprime loans do not face these limits. As a result, borrowers choosing subprime loans rather than FHA loans were often doing so to reduce monthly payments or to finance more expensive homes using alternative financing features, rather than to relax downpayment constraints.

Figure A1: Servicer Coverage Compared to HMDA



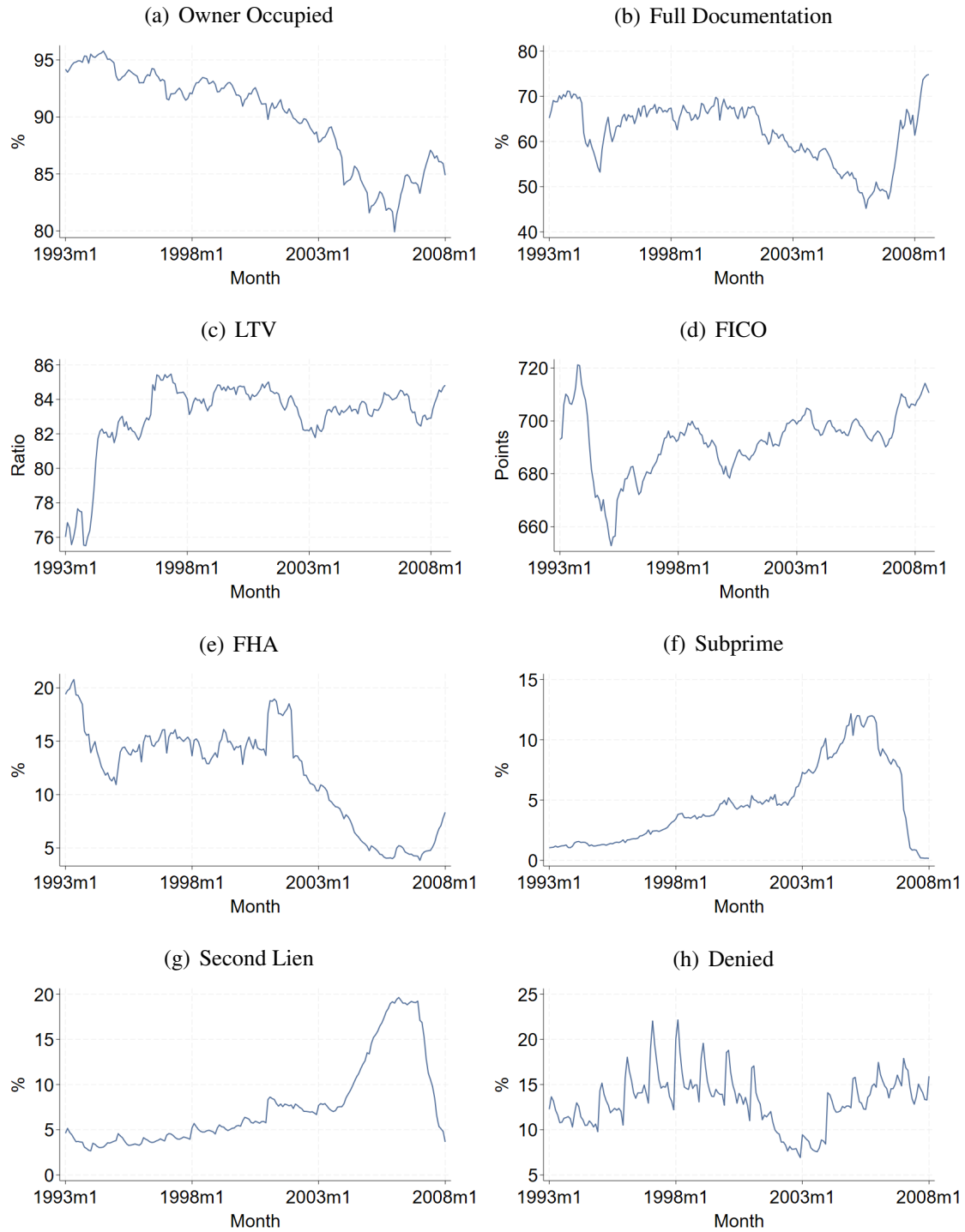
Note: This figures shows the count and volume of purchase first-liens in the McDash LPS and CL servicer data as compared to the HMDA data.

Figure A2: National Trends in Interest-modifying and Non-interest-modifying Features



Note: This figure shows the share of purchase mortgages with an interest-modifying alternative feature (interest only, negative amortization, or teaser) versus a non-interest-modifying alternative feature (extended term or balloon) as well as both types of features.

Figure A3: National Trends in Other Mortgage Attributes



Note: The figure shows the time series for features of new purchase loan originations not related to alternative financing features.

6.B Payment Gap Measurement

Table A1: Determinants of Mortgage Rate Spread Over Treasury Rates

Variable	Coefficient	Standard Error
FICO 2	.027	.053
FICO 3	-.019	.04
FICO 4	-.129	.039
FICO 5	-.264	.034
LTV 2	.021	.039
LTV 3	.004	.038
LTV 4	-.075	.033
LTV 5	-.027	.037
FICO 2 x LTV 2	-.070	.064
FICO 2 x LTV 3	-.023	.062
FICO 2 x LTV 4	-.044	.055
FICO 2 x LTV 5	-.044	.062
FICO 3 x LTV 2	.051	.049
FICO 3 x LTV 3	.047	.048
FICO 3 x LTV 4	-.017	.042
FICO 3 x LTV 5	-.067	.049
FICO 4 x LTV 2	.081	.049
FICO 4 x LTV 3	.176	.049
FICO 4 x LTV 4	.020	.042
FICO 4 x LTV 5	-.021	.052
FICO 5 x LTV 2	.048	.042
FICO 5 x LTV 3	.175	.042
FICO 5 x LTV 4	.065	.036
FICO 5 x LTV 5	.061	.045
R-Sq	.019	
Within R-Sq	.019	
No. Obs	53,757	
No. Counties	2,082	

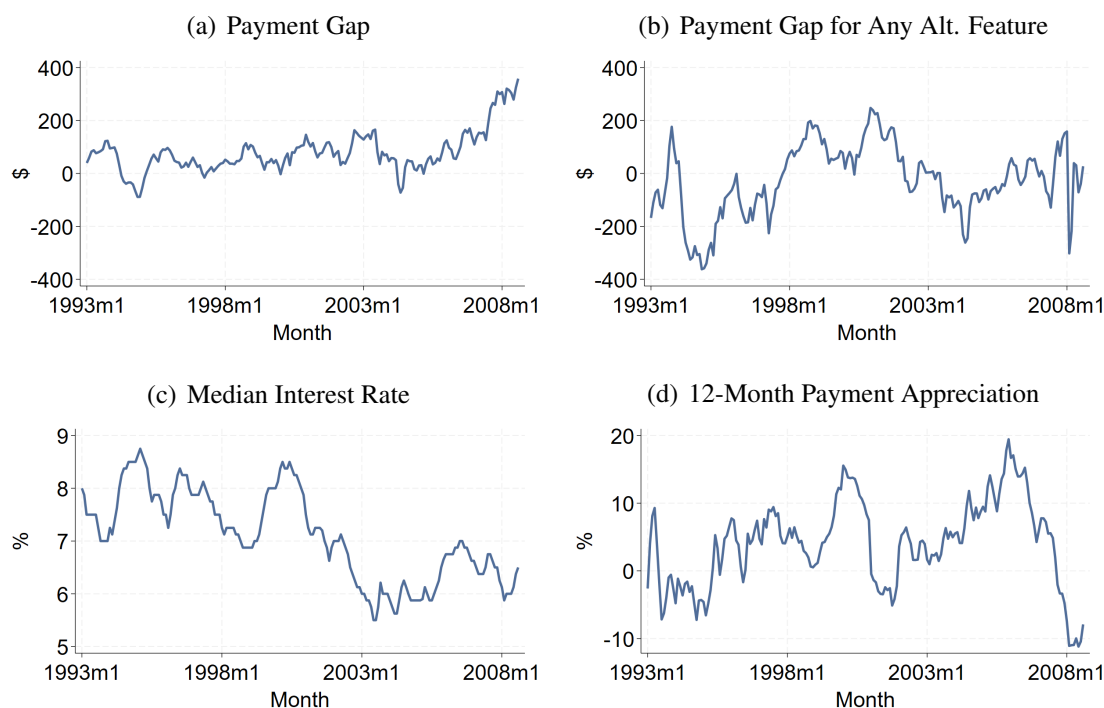
Note: This table presents the results from the estimation of equation 1. The spread is the difference between the borrower's interest rate at origination and the 10-year Treasury rate from the month of origination, limited to originations for 30-year FRMs in 1995. As expected borrowers in lower FICO and higher LTV buckets generally show a higher spread. FICO bucket boundaries defined at scores of 580, 620, 680, and 720. LTV bucket boundaries defined at ratios of 70, 80, 90, and 100.

Table A1 reports the estimates used to construct the counterfactual mortgage rates underlying the payment gap measure described in Section 2.A. As discussed in the main text, we first estimate the relationship between the spread on mortgage contract rates over the 10-year Treasury rate and key loan characteristics using 30-year fixed-rate mortgages originated in 1995 (equation 1). The specification allows the spread to vary across borrower credit quality and leverage through indicators for FICO and LTV buckets and their interactions. The estimated coefficients from this

regression are then combined with the path of Treasury rates to project counterfactual 30-year FRM interest rates for loans originated throughout the sample period using equation 2. These projected rates form the basis for the counterfactual payments used to measure the payment gap.

Figure A4 shows the evolution of our aggregate payment gap measure over time alongside other summary features of mortgage payments. Panel (a) shows the median payment gap for all loans and panel (b) shows the median payment gap conditional on loans with an alternative financing feature. Throughout the late 1990s, we estimate a negative payment gap for about half of mortgages. The median payment gap declines by \$47 from January 2002 to April 2005, which is less than the \$118 decline among mortgages with alternative financing features. This difference is because the aggregate measure incorporates payment increases from traditional mortgage products.

Figure A4: Payments



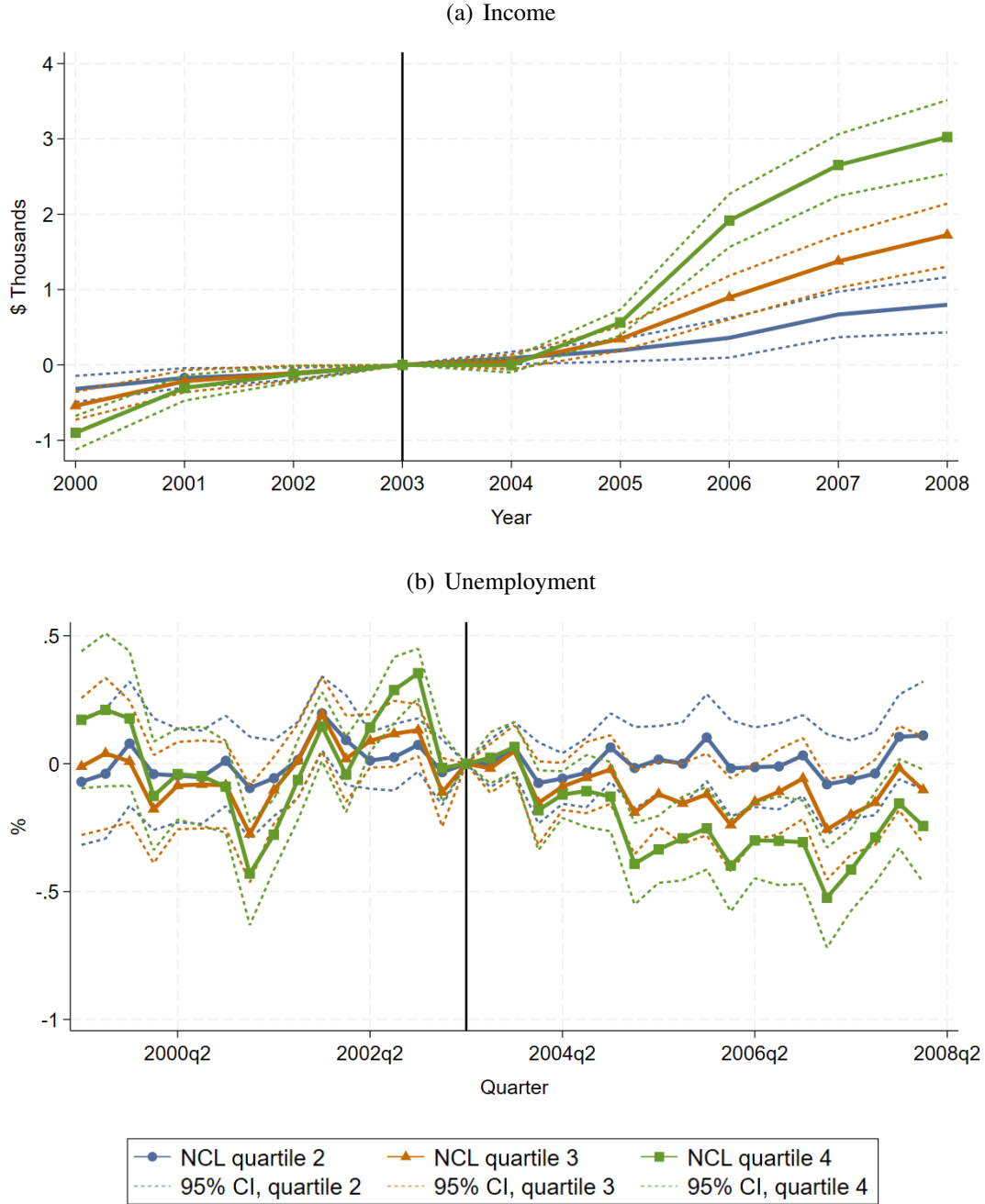
Note: This figures shows aggregate payment features. Panel (a) shows the median payment gap for all purchase loans while panel (b) shows the median payment gap for loans with any alternative financing feature. Panel (c) shows the median interest rate at origination. Panel (d) shows 12-month growth in the median payment.

The bottom two panels of Figure A4 further emphasize the experienced mortgage savings of borrowers during the housing boom. In panel (c) we show the median interest rates on new mortgages. Consistent with Justiniano, Primiceri and Tambalotti (2022), we show that the median interest rate on mortgages increased little from mid-2003 through the end of the boom, even while Treasury rates steadily rose. Finally, in panel (d) we show median payment appreciation for new

mortgages. While median payment appreciation during the boom did steadily increase, the rate of appreciation lagged house price appreciation by 5 percentage points.

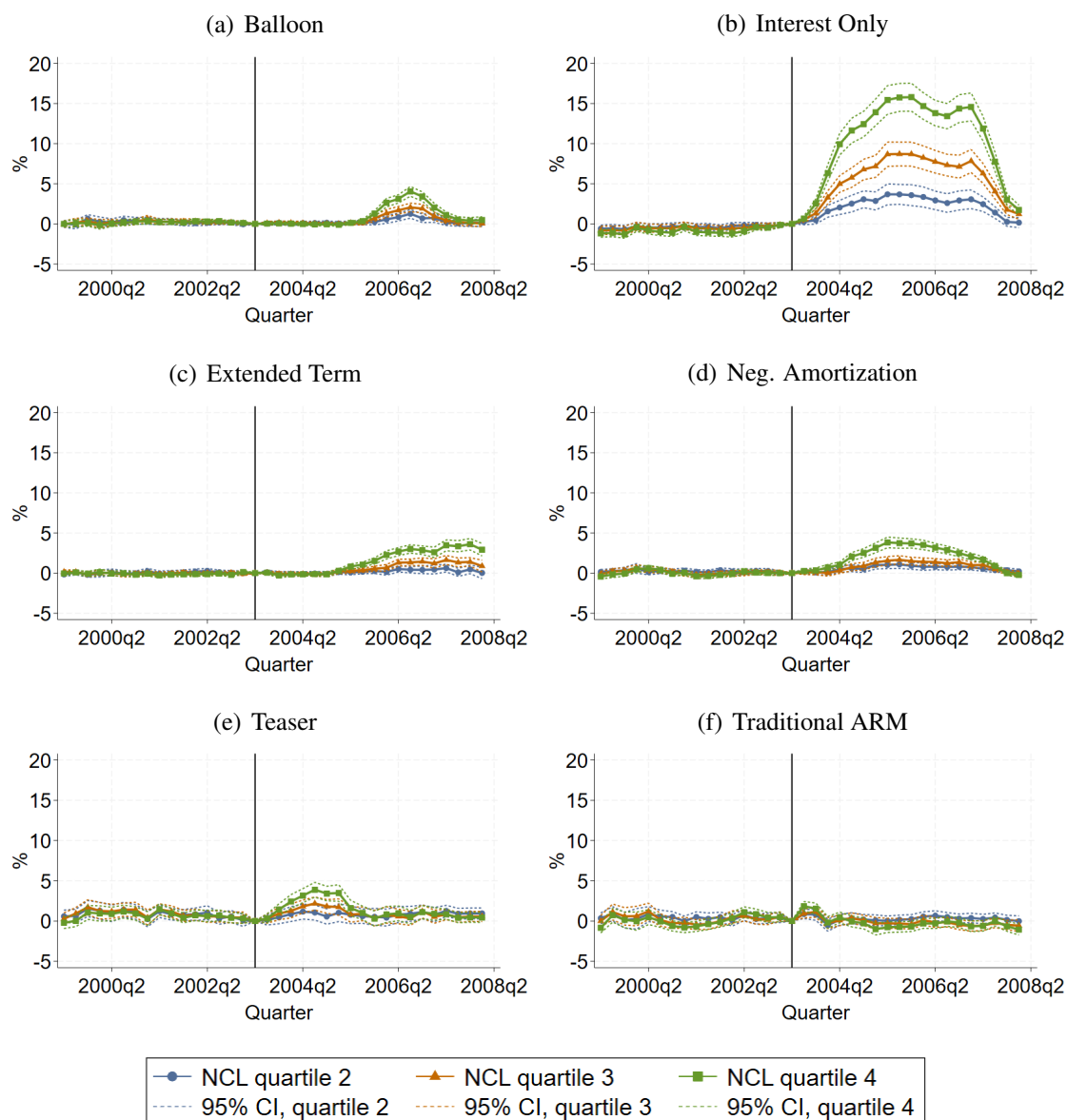
6.C Additional Results: PLS Market Event Study

Figure A5: Economic Fundamentals by Non-core Lending Quartile



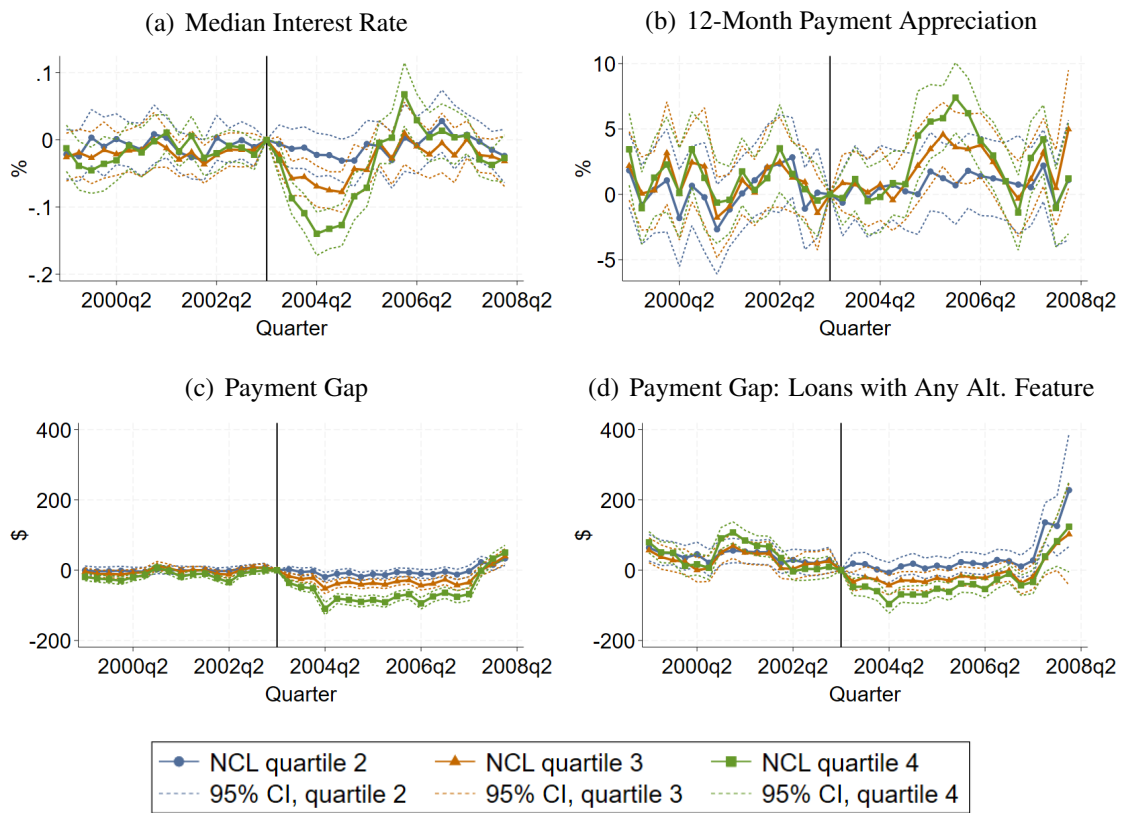
Note: This figure shows income and unemployment rate changes around the mid-2003 acceleration of the private label securities market. Counties are divided into quartiles of exposure to non-core liability lenders. The coefficients are derived via equation 5 and are relative to the first (bottom) quartile of counties.

Figure A6: Alternative Financing Features by Non-core Lending Quartile with Additional Controls



Note: This figure shows the use of alternative mortgage features around the mid-2003 acceleration of the private label securities market. Changes in use of non-teaser ARMs are shown for comparison. Counties are divided into quartiles of exposure to non-core liability lenders. The series shown are relative to the first (bottom) quartile of non-core deposits.

Figure A7: Payments by Non-core Lending Quartile with Additional Controls



Note: This figure describes changes in payments around the mid-2003 acceleration of the private label securities market. Changes in use of non-teaser ARMs are shown for comparison. Counties are divided into quartiles of exposure to non-core liability lenders. The series shown are relative to the first (bottom) quartile of non-core deposits.

Table A2: OLS Estimates for Changes in House Price Appreciation and Credit Supply

Panel A: Alternative Financing Features

	Balloon (1)	Interest Only (2)	Extended Term (3)	Neg. Amortization (4)	Teaser (5)	Traditional ARM (6)
HPA Percent Change 2002m1 to 2005m4	0.009 (0.007)	0.132*** (0.006)	0.037*** (0.007)	0.134*** (0.006)	0.017** (0.007)	-0.007 (0.007)
Mean Change	-1.026	14.363	0.791	3.244	15.494	0.085
Std Deviation Change	2.576	14.191	3.297	5.503	10.536	6.916
N	1156	1156	1156	1156	1156	1156
R-Squared	0.001	0.317	0.025	0.328	0.005	0.001

Panel B: Payments

	Pmt. Gap (1)	Pmt. Gap, Any (2)	Med. Interest Rate (3)	12-Month Pmt. Appr. (4)
HPA Percent Change 2002m1 to 2005m4	-0.070*** (0.007)	-0.050*** (0.008)	-0.038*** (0.007)	0.149*** (0.005)
Mean Change	11.646	-38.849	-0.907	4.696
Std Deviation Change	115.198	240.641	0.319	7.669
N	1150	879	1155	1162
R-Squared	0.090	0.042	0.027	0.407

Panel C: Other Mortgage Attributes

	Owner Occupied (1)	Full Documentation (2)	LTV (3)	FICO (4)
HPA Percent Change 2002m1 to 2005m4	-0.065*** (0.007)	-0.031*** (0.007)	-0.043*** (0.007)	0.031*** (0.007)
Mean Change	-4.295	-0.670	0.930	0.807
Std Deviation Change	8.325	26.219	5.430	26.210
N	1162	1097	1156	1151
R-Squared	0.077	0.017	0.033	0.018

Panel C (Cont.): Other Mortgage Attributes

	FHA (1)	Subprime (2)	Second Lien (3)	Denied (4)
HPA Percent Change 2002m1 to 2005m4	-0.038*** (0.007)	0.048*** (0.007)	0.038*** (0.007)	0.043*** (0.007)
Mean Change	-6.223	4.957	5.029	1.816
Std Deviation Change	7.719	5.624	6.610	6.307
N	1162	1162	1162	1162
R-Squared	0.026	0.041	0.026	0.033

Note: Significance levels denoted by * 0.10 ** 0.05 *** 0.01. This table presents the uninstrumented OLS results from equation 4. House price appreciation and each measure of credit supply are standardized by their mean and standard deviation, shown as a summary statistic for each regression.

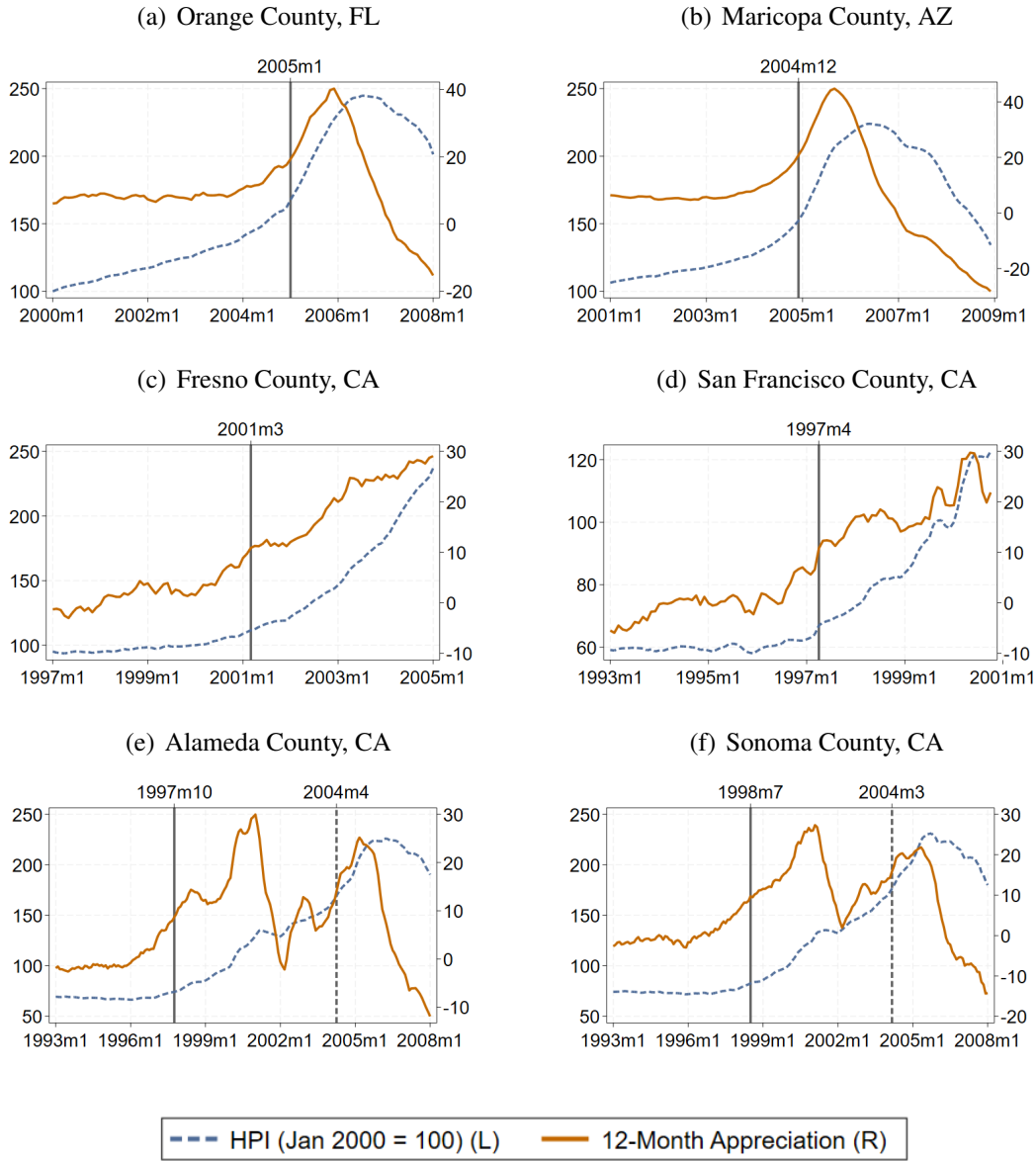
6.D Breakpoint Methodology

An important caveat to our application of standard breakpoint methodology is dictated by the data. Other breakpoint studies typically split their sample into two parts: one that they use to estimate the breakpoint, and another to estimate the size of the jump (e.g. [Card, Mas and Rothstein 2011](#)). In doing so, some issues related to specification search bias can be ameliorated ([Leamer 1983](#)). We are unable to do this because we take our house price indices as given, rather than calculated directly from microdata. Our analysis applies already-constructed house price indices from CoreLogic because we would otherwise need either: (1) the exact property address to construct our own repeat-sales index or (2) detailed property attributes (such as square footage, number of bedrooms) to estimate a hedonic index. Because we have neither, we cannot construct a credible house price series with our loan-level data. To gauge how our conclusions might be affected by this issue, we conduct a robustness analysis where we weight our analysis using the t-statistic for λ^{**} that is obtained when estimating Equation (16). We also compare the distribution of our house price breaks to those found in studies that apply a split sample approach (e.g. [DeFusco et al. 2018](#); [Ferreira and Gyourko 2023](#)). These analyses suggest that our conclusions would be little changed if we were to apply a split sample approach, and these results are available upon request.

While our structural break approach is the most popular one used to identify interruptions in housing market trends, we note that two other approaches for estimating breakpoints yield largely similar results. In results not shown, we examined the sup-Wald test, which extends the Chow test to cases with unknown breakpoints as in [Andrews \(1993\)](#). This approach searches for the maximum Wald statistic (rather than the sum of squared errors) from a nearly identical specification to [Hansen \(2000\)](#). In contrast, the indicator saturation method starts with a general model with a breakpoint in every time period and then conducts model selection algorithmically to arrive at significant breaks in the time series (through step-saturation, see [Doornik, Hendry and Pretis \(2013\)](#)). Selecting the statistically significant break with the largest t-statistic yields breaks that are very highly correlated. Both alternative approaches produce county-level breaks that are usually within 6 months of the [Hansen \(2000\)](#) break.

Figure [A8](#) provides examples of how our chosen breakpoint methodology identifies housing boom starts for counties with differently shaped house price paths. In the top row, Orange County, FL and Maricopa County, AZ have a constant rate of house price appreciation until the mid-2000s, when house prices accelerate into a boom. Our estimated breakpoint for each county aligns with this acceleration and chooses precisely which month in that period is best characterized as the start of the boom. In the second row, Fresno County, CA and San Francisco County, CA do not have visually clear breaks in their paths of house price appreciation. However, our methodology finds that there is a statistically significant break in these two counties. Finally, in the last row we show

Figure A8: House Price Breaks for Sample Counties



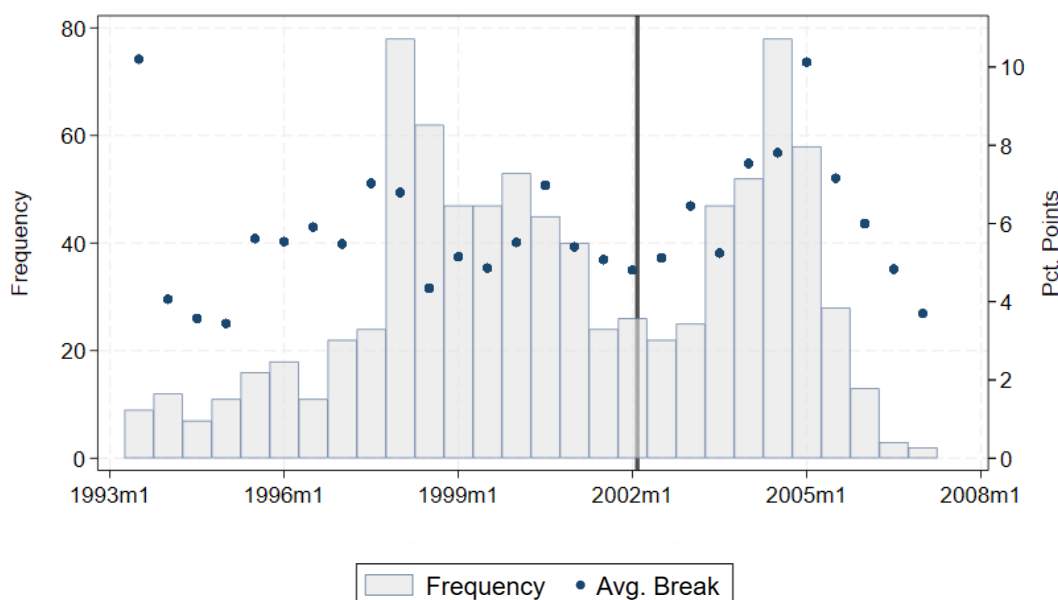
Note: This figure shows the timing of estimated structural house price breaks for three typical house price paths. The first row shows counties with one visually clear boom in house prices. The second row shows counties with accelerating house prices with visually ambiguous starts to a boom. The third row shows counties that are estimated to have two distinct booms. Each panel shows both the house price index and 12-month appreciation rate.

two counties, Alameda County, CA and Sonoma County, CA, for which the path of house price appreciation suggests the possibility of two booms. Indeed, conditional on the statistical detection of initial breakpoints for both counties in the late 1990s, both counties experience a statistically significant second break in 2004. In our main analysis we focus on counties' first breakpoints and

find that the inclusion of these second breakpoints has no qualitative impact on our main results in the robustness section.

At the county level, we find substantial heterogeneity in the timing and size of house price booms across the counties in our sample. Figure A9 shows a histogram of the timing and size of house price breaks. The majority of the structural breaks occur between 1998 and 2006 with two clear clusters. The earlier cluster around 1998 is associated with the tech boom in Coastal California and the Northeast. Another surge in local house price breaks occurs in 2004, aligning with the acceleration in the PLS market. Figure A9 also shows that there was wide variation in the average magnitude of the house price break over time, ranging from 3.4 to 10.2 percentage points.

Figure A9: Distribution of House Price Breaks with Average Magnitude



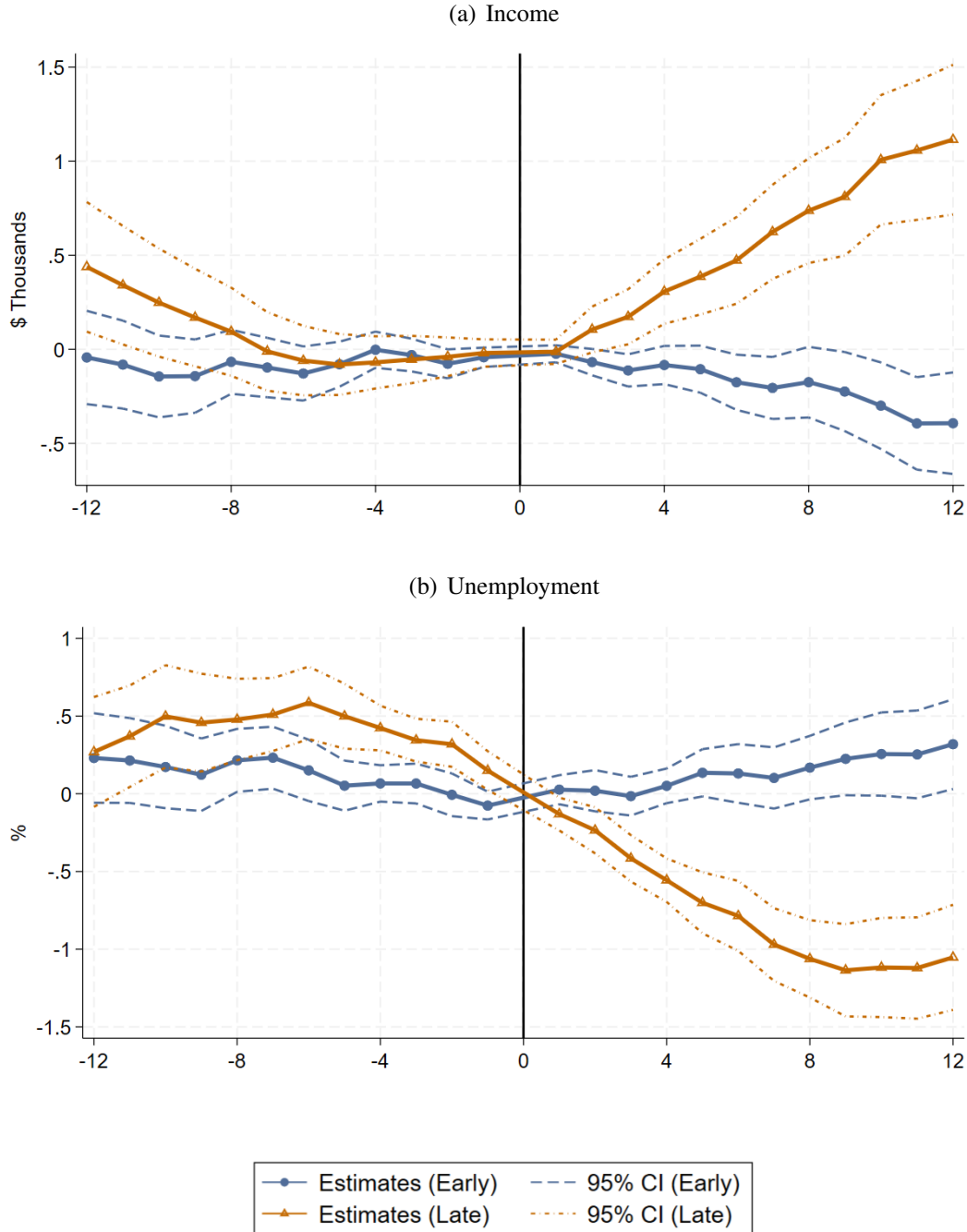
Note: This figures shows the frequency distribution of house price breaks across time and the average size of the booms that start in each month. We estimate 984 counties with a valid house price break. The magnitude of house price break is calculated as $HPA_{t+11} - HPA_{t-1}$, where t is the month of the first house price break. Vertical line indicates February 2002. Breaks before the line are classified as early, while breaks after the line are classified as late.

6.E Additional Results: Early vs. Late Housing Booms

6.E.1 Economic Fundamentals

We also examine household income and unemployment around the start of local housing booms using a modification of equation 8 that excludes these variables as controls. Figure A10 shows little evidence that improving economic fundamentals precede the start of either early or late housing booms. In late-booming markets unemployment improves modestly in the four quarters before the boom while income remains largely steady. After the start economic conditions diverge across early and late booms. In early-booming markets income and employment conditions deteriorate two to four years later which reflects the late 1990s dot-com bust. In contrast both income and unemployment improve after the start of late booms. [Ferreira and Gyourko \(2023\)](#) similarly find no income pre-trends in the earliest markets but rising incomes around the start of mid 1990s housing booms.

Figure A10: Economic Fundamentals around House Price Breaks



Note: This figure shows the event study figures for economic fundamentals. The start of the the housing boom is $t = 0$ (defined as the quarter of the structural break in house prices). Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals. We show a yearly event student for house price appreciation because the start of local housing booms are defined using year-of-year appreciation. For these outcomes, our specification only includes county fixed effects and a quadratic in calendar time for equation 8.

6.E.2 Incorporating Second Breaks

The patterns for counties such as Alameda County, CA and Sonoma County, CA in Figure A8 suggest that limiting our methodology to one breakpoint between 1993 and the height of the county-specific house price appreciation misses secondary booms that may also be informative. To incorporate these booms, we repeat our methodology for identifying first booms, but restrict the estimation of equations 6 and 7 to the period between the start of the first boom and the month of peak house price appreciation after the first boom. We further require second breaks to occur at least 12 months after the first. We find that of the 984 counties with at least one structural break in house prices, 452 counties also have a second.

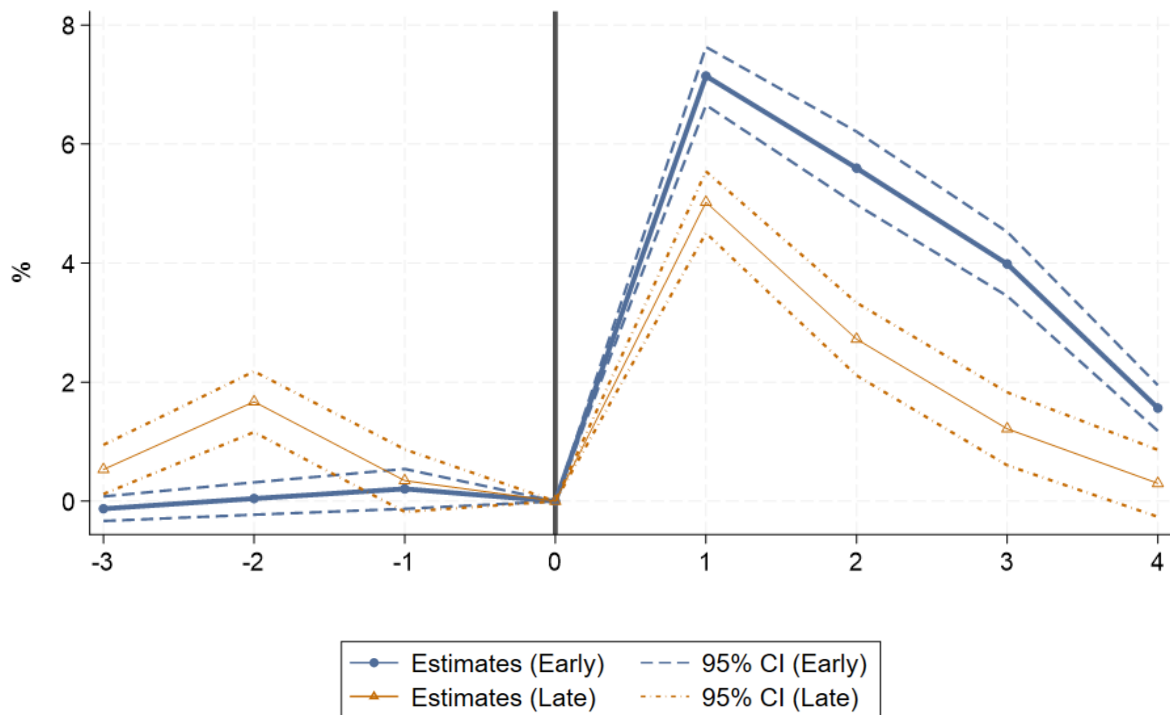
There are some distinct differences between first and second breaks. Figure A11 shows the histogram of the timing and size of these second breaks. Second house price breaks cluster quite late over the period we study; 28 percent occur between mid-2004 and mid-2005, when many other counties experience the start of their first house price boom around the expansion of the PLS market. Figure A12 presents the full path of house price appreciation in the years before and after the booms, allowing the paths to differ by whether the boom is the first or the second. Counties with two house price breaks tend to experience breaks of smaller magnitude overall, with the second slightly smaller than the first. Thus, conditions in the 2000s produced substantial housing booms — even in counties with a history of recent housing booms. When we study non-traditional mortgage use around pooled first and second breaks in Figures A13 and A14, we find our main results to be qualitatively and quantitatively similar.

Figure A11: Distribution of First and Second House Price Breaks with Average Magnitude



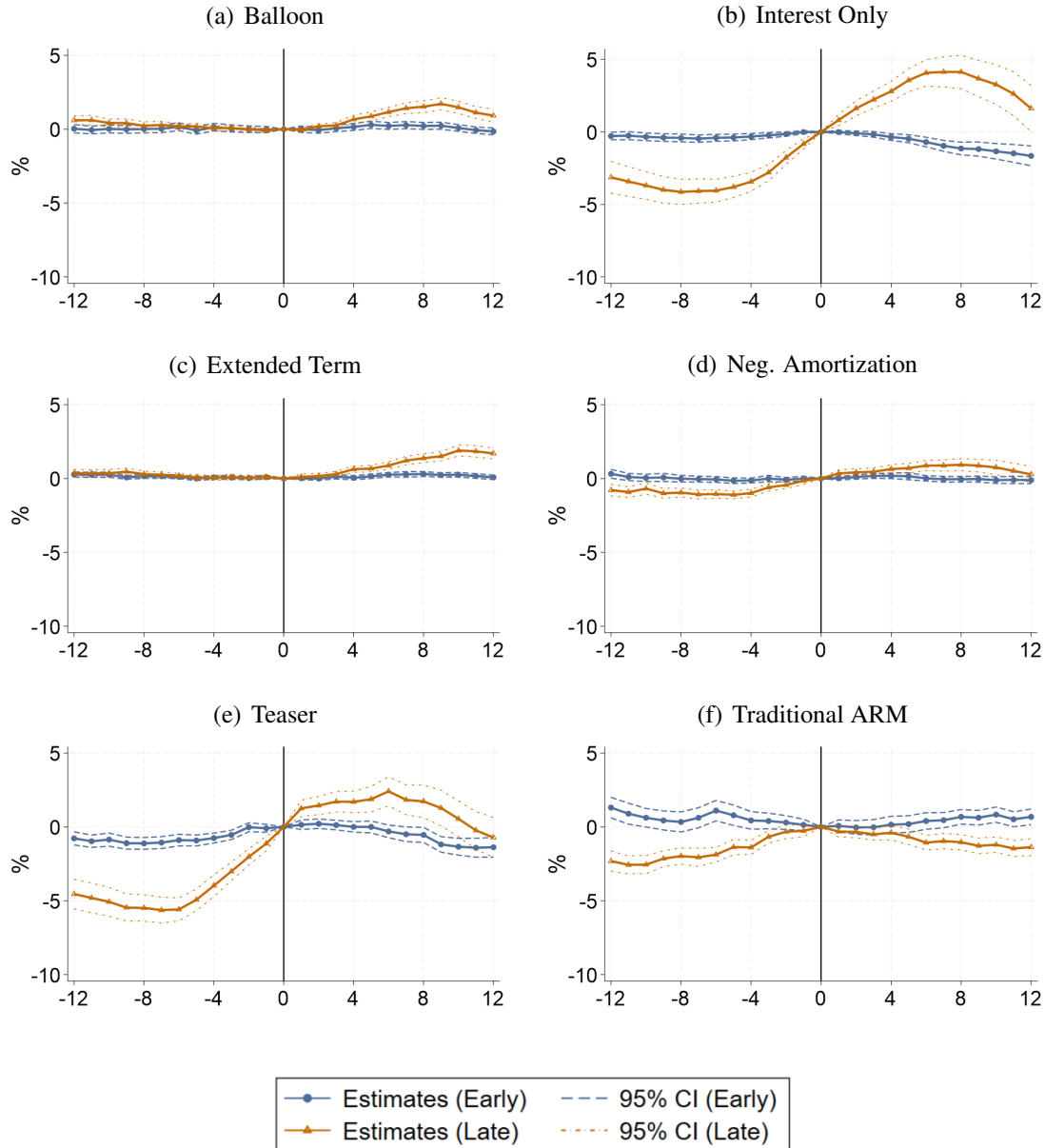
Note: This figure shows the distribution of first and second house price breaks. 984 counties have a valid first house price break. 452 counties have a valid second house price break. The magnitude of house price break is calculated as $HPA_{t+11} - HPA_{t-1}$, where t is the month of the house price break.

Figure A12: House Price Appreciation around First and Second House Price Breaks



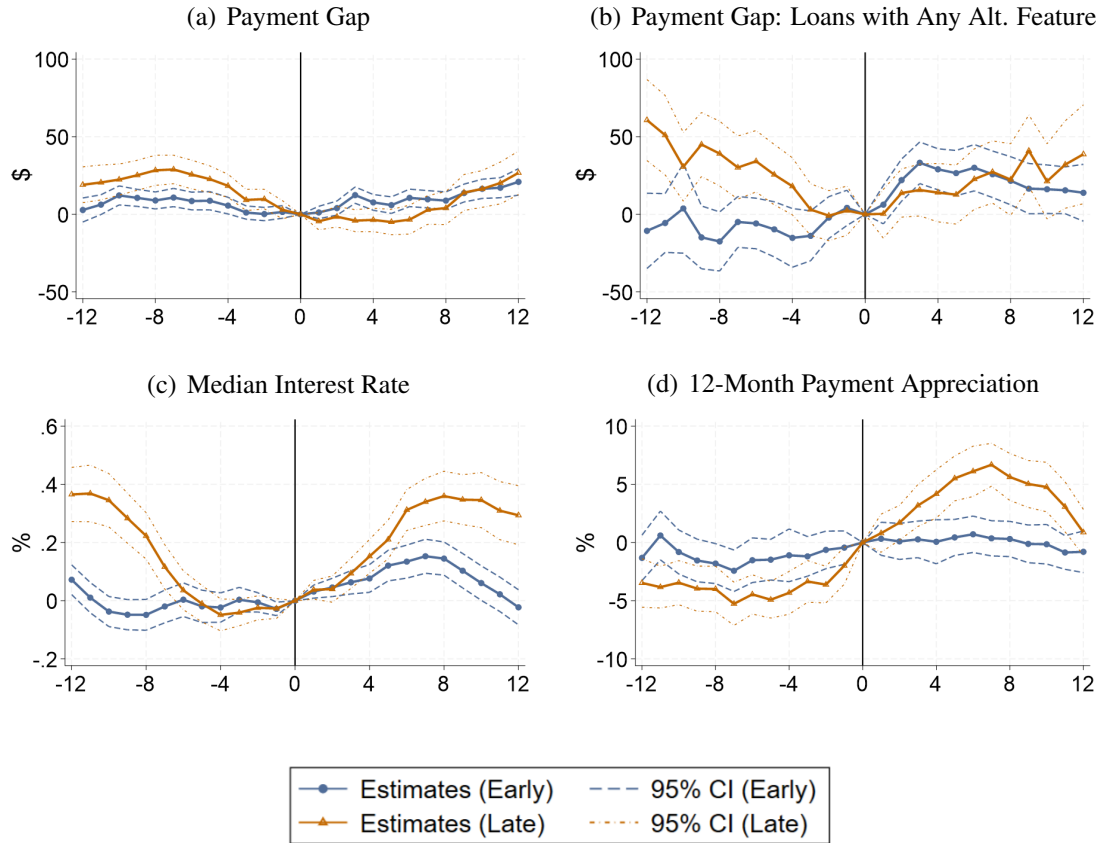
Note: This figure shows yearly event study results for house price appreciation around counties first and second house price breaks. Counties are defined to have a valid first break if the corresponding break is statistically significant at 95% and have at least 36 months between January 1993 and the month of peak HPA. Similarly, counties are defined to have a valid second break if the corresponding break is statistically significant at the 95% and have at least 12 months between the first peak and the second peak. Among 1,163 in-sample counties, 984 have a valid first house break and 452 have a valid second house break.

Figure A13: Alternative Financing Features around House Price Breaks, Including Second Breaks



Note: This figure presents the event study results for alternative financing features where $t = 0$ is the quarter of the structural break in house prices, including both first and second house price breaks. Event study specification include county fixed effects, a quadratic in calendar time and economic fundamentals (income and unemployment). Dashed lines represent 95 percent confidence intervals. Early breaks are before February 2002 and late breaks are February 2002 or later.

Figure A14: Payments around House Price Breaks, Including Second Breaks



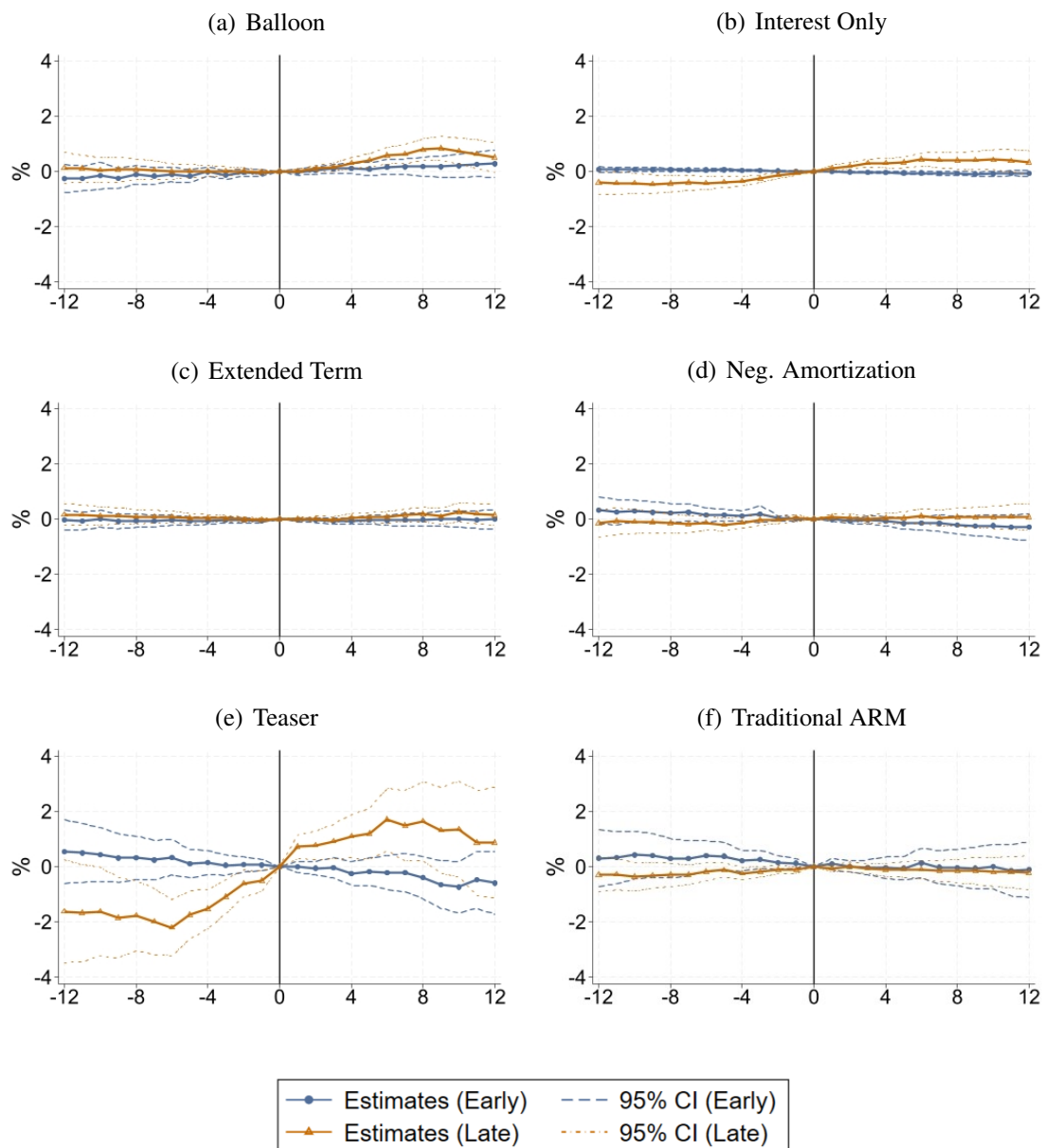
Note: This figure presents event study results for alternative financing features where $t = 0$ is the quarter of the structural break in house prices. The event study specification include county fixed effects, a quadratic in calendar time and economic fundamentals (income and unemployment). Early counties are defined to have a first house break before February 2002 and late counties have a break in February 2002 or later. Dashed lines represent 95 percent confidence intervals.

6.E.3 Non-traditional Product Use by FICO

[Amromin et al. \(2018\)](#) use the Black Knight McDash data to study the borrower profile of what they call complex mortgage products. Unlike our study, they define complex mortgages more narrowly to include only interest-only and negative-amortization products. They find that borrowers of more complex products were more likely to be of prime quality. Given that we find increases in non-traditional mortgage products coincident with increases in subprime lending prior to increases in house prices, this raises questions around the use of non-traditional products by subprime borrowers. First, with servicer data from both Black Knight McDash and CL, we may cover a portion of the market in which subprime borrowers tended to use interest-only and negative-amortization products more. Second, the use of non-traditional products specifically by subprime borrowers might be a force driving the start of housing booms without subprime borrowers dominating the market for non-traditional products overall. Third, both prime and subprime borrowers may have preferred different types of non-traditional products with increased use of different products by both segments prior to the start of local housing booms.

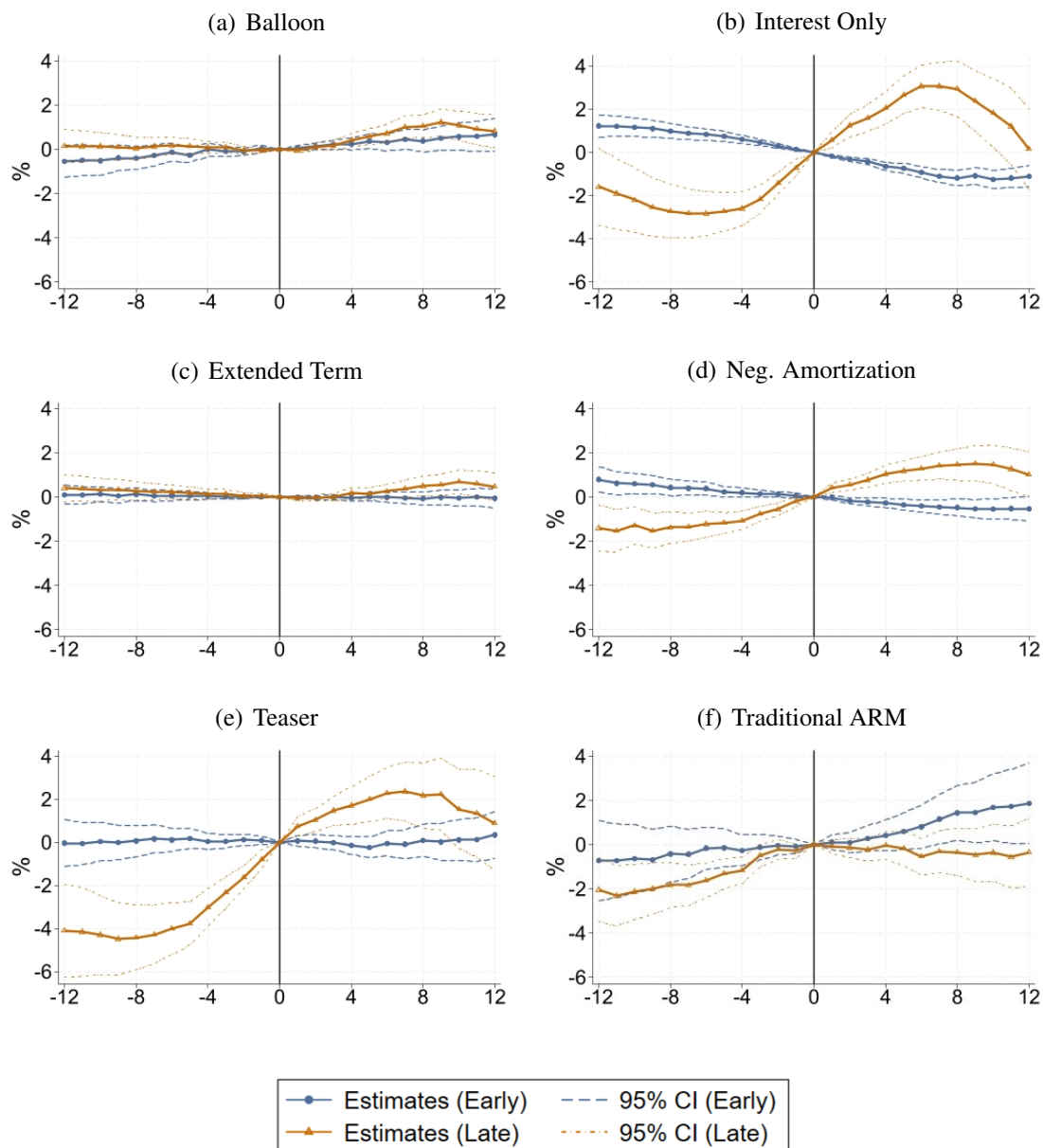
We find this third explanation to be the most plausible. Although our subprime and alternative features measures by county and month are derived from different datasets, we can use the servicer data to separate the use of alternative features by FICO score. Using a common definition of subprime as borrowers with a FICO score under 620, we re-estimate our main specification with this alternative definition by subprime and non-subprime borrowers. These results are shown in Figures [A15](#) and [A16](#). We find that subprime borrowers substantially increased their use primarily of teaser-rate loans. In contrast, borrowers with FICO scores over 620 show sharp jumps in the use of all non-traditional mortgages with an interest-modifying feature as well as more traditional ARMs prior to the start of late booms. We view these results as providing new and expanded support for the affordable product borrower profile described by [Amromin et al. \(2018\)](#).

Figure A15: Alternative Financing Features around House Price Breaks for FICO below 620



Note: This figure presents the event study results for the use of alternative financing features by borrowers with FICO scores below 620, where $t = 0$ is the quarter of the structural break in house prices. Event study specification include county fixed effects, a quadratic in calendar time and economic fundamentals (income and unemployment). Dashed lines represent 95 percent confidence intervals. Early counties have their first house break before February 2002 and late counties have their first break in February 2002 or later.

Figure A16: Alternative Financing Features around House Price Breaks for FICO at least 620



Note: This figure shows the event study results for the use of alternative financing features by borrowers with FICO scores of at least 620, where $t = 0$ is the quarter of the structural break in house prices. Event study specification includes county fixed effects, a quadratic in calendar time and economic fundamentals (income and unemployment). Dashed lines represent 95 percent confidence intervals. Early counties have their first house break before February 2002 and late counties have their first break in February 2002 or later.

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